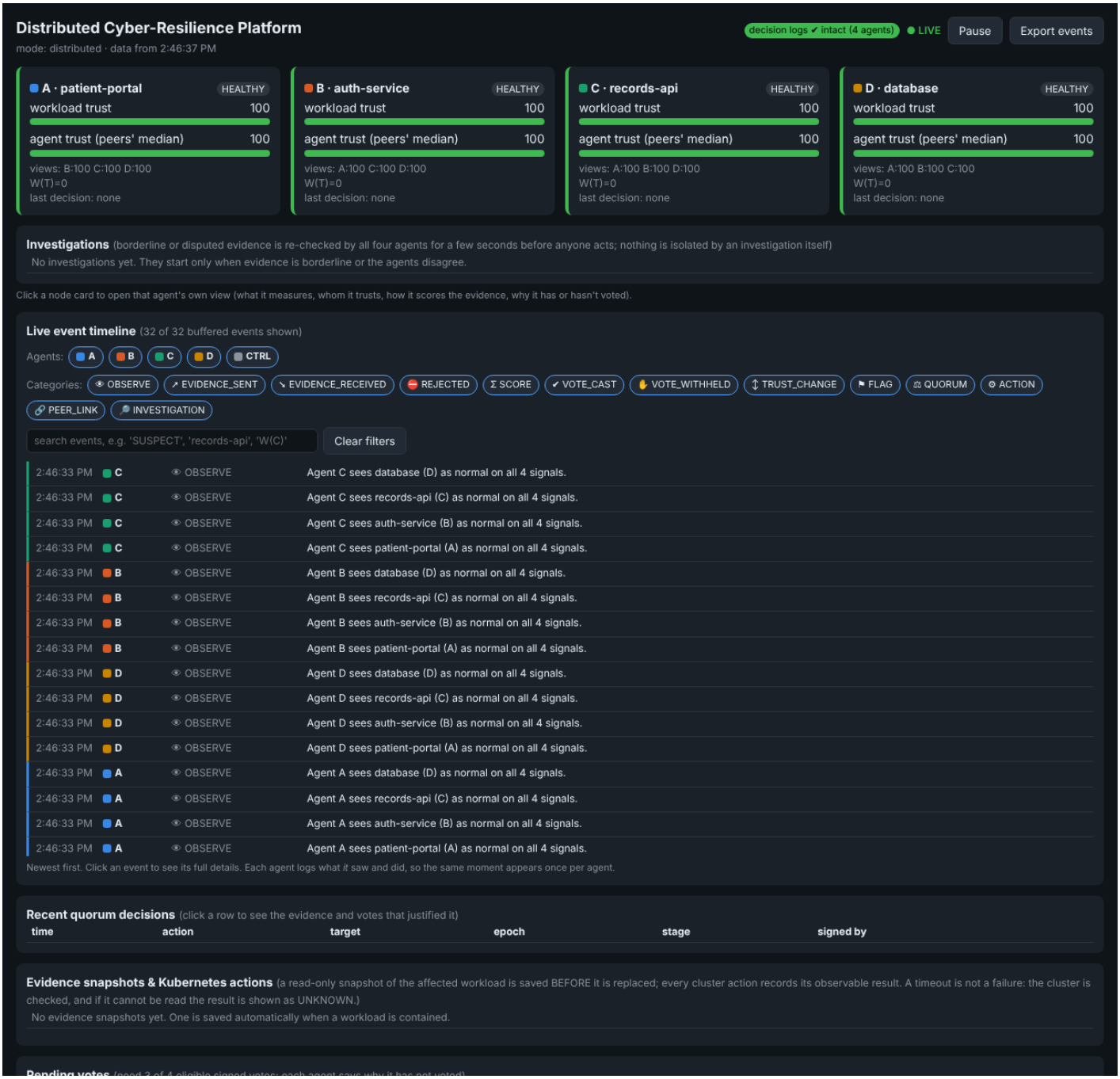
Distributed Cyber-Resilience Platform

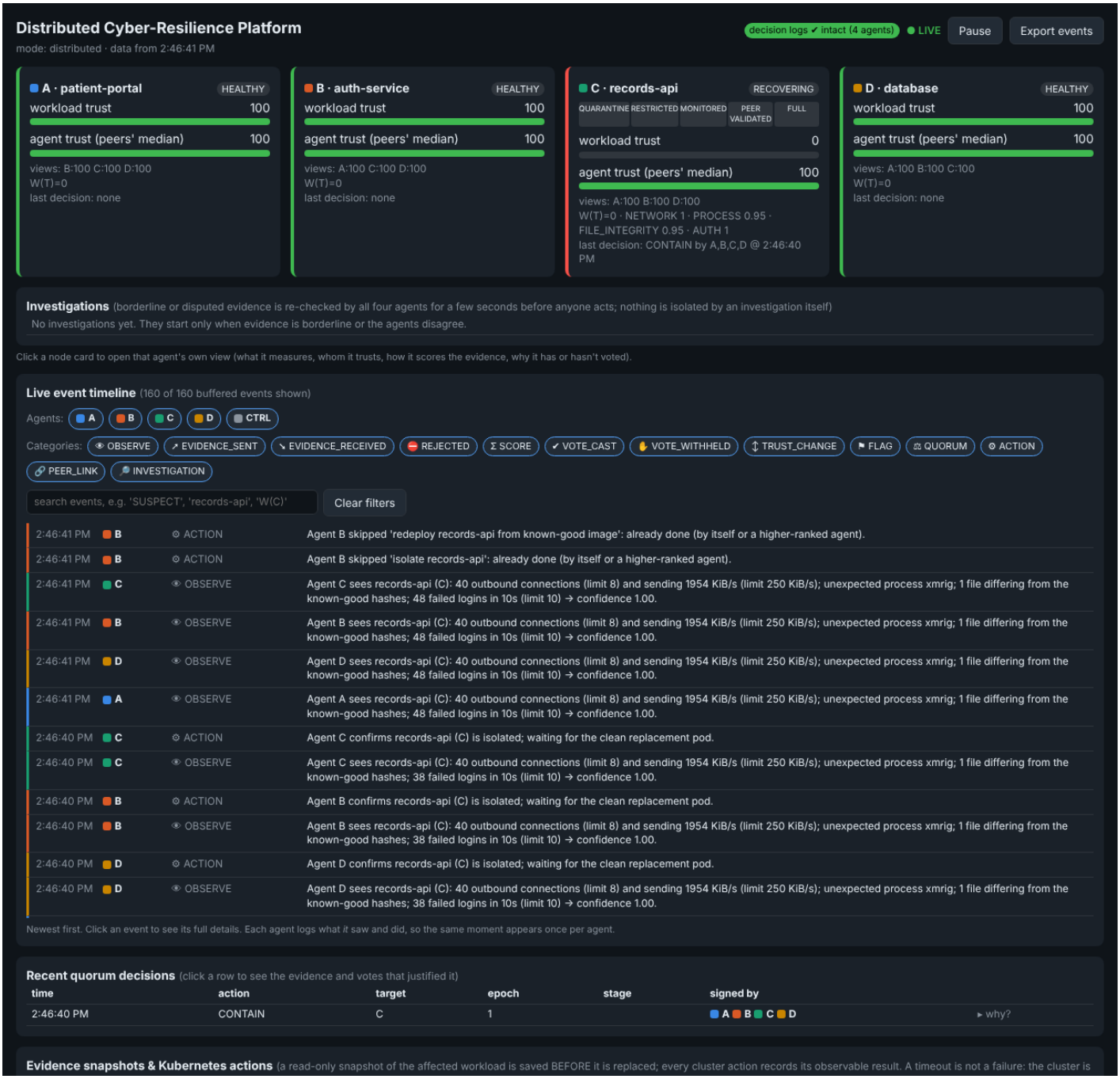
Dashboard screenshots, from the early phases to Healthcare Phase D

All screenshots were taken from the SIMULATOR (simulated telemetry and Kubernetes API; real agents, quorum and certificate
or from test fixtures where stated. None was taken on a real cluster. All patient data is synthetic.
The connected Live Monitoring page is not shown: it can only be captured on your own cluster.
Source files: docs/img/ in the repository (full-size originals).

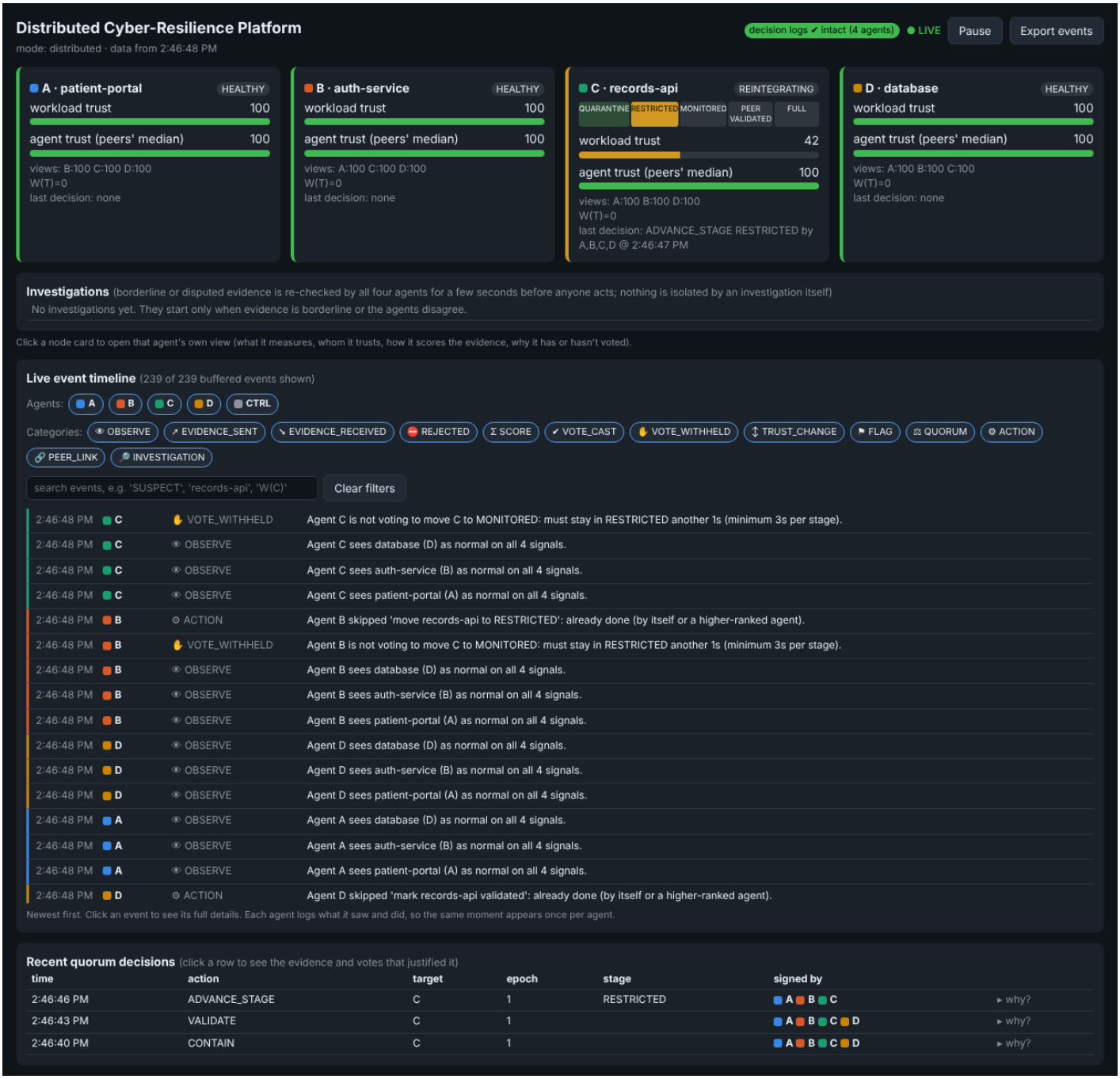
Early phases: healthy state (4 workloads, 4 agents)



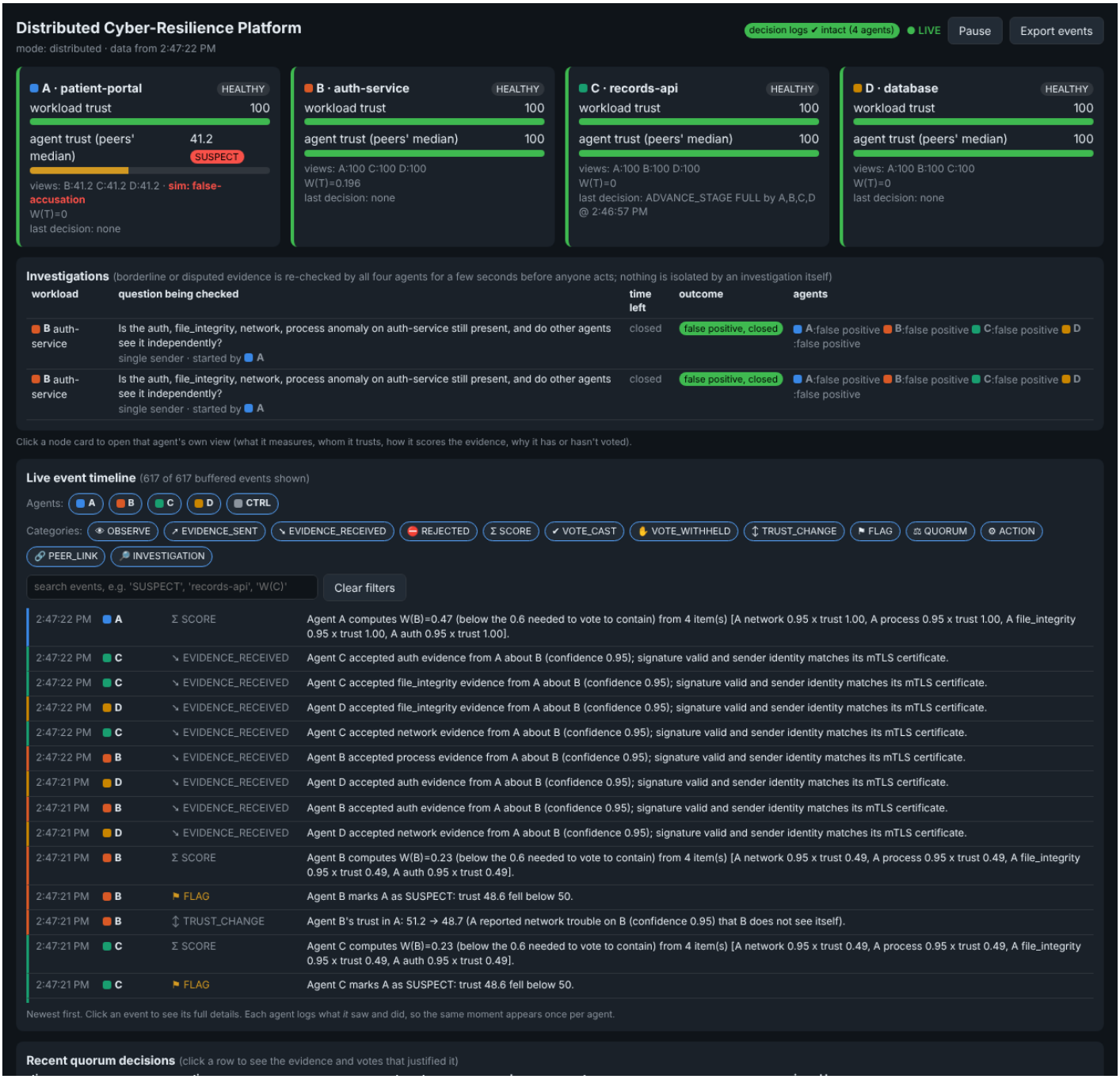
Early phases: a genuine compromise contained by the quorum



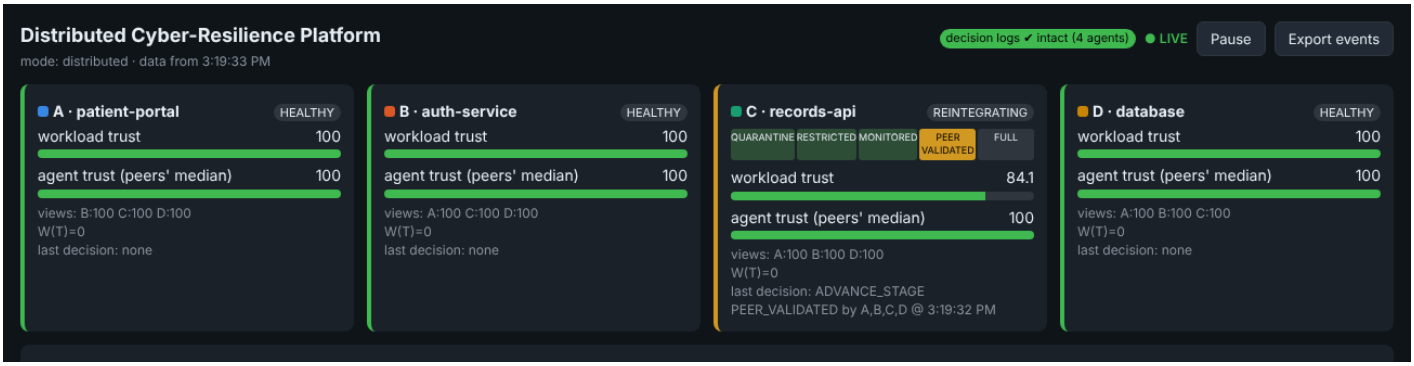
Early phases: staged return to normal access



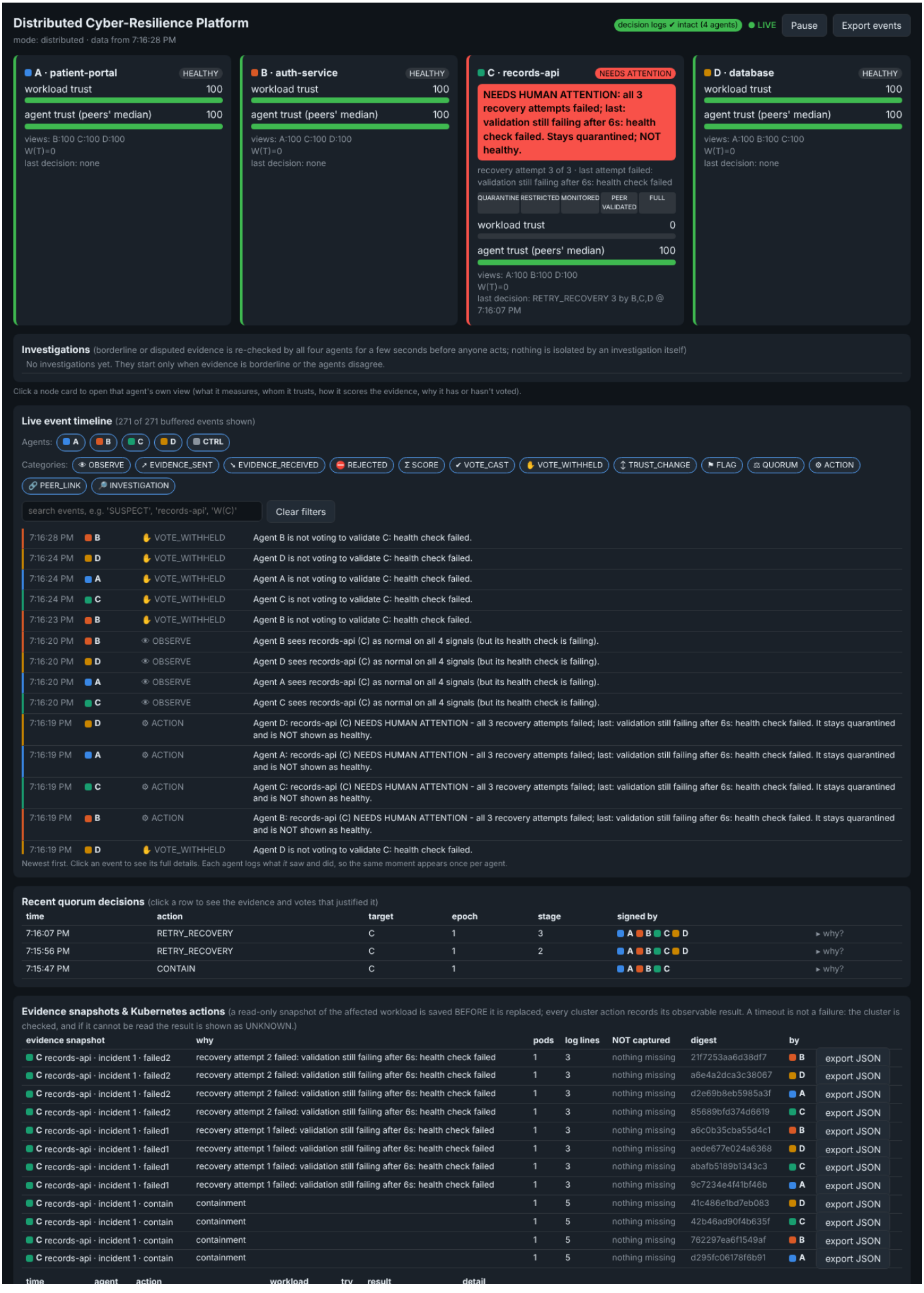
Early phases: a lying agent is flagged, no false isolation



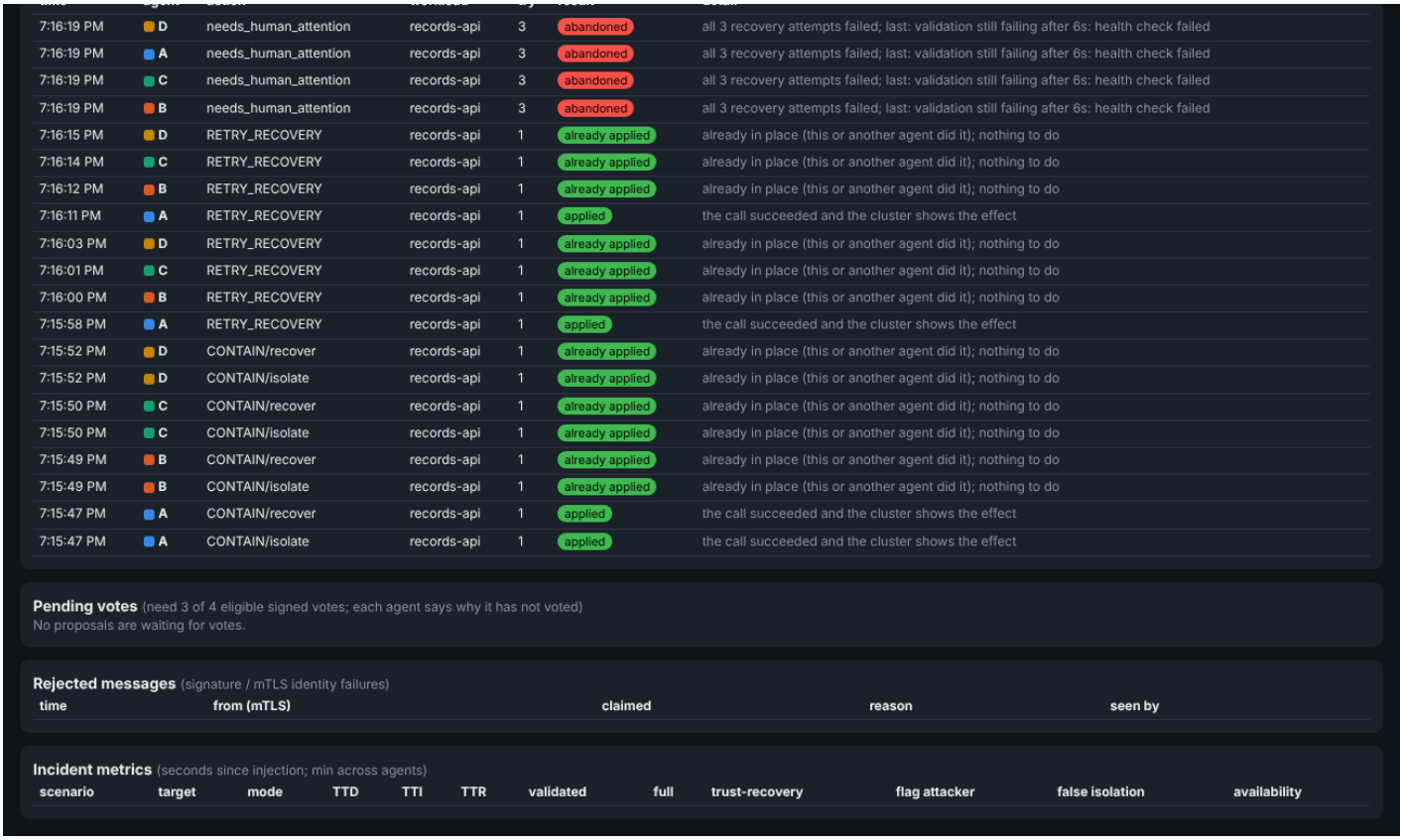
Early phases: decision-log integrity chip



Early phases: recovery failed - needs human attention (part 1/2)



Early phases: recovery failed - needs human attention (part 2/2)



7:16:19 PM	D	needs_human_attention	records-api	3	abandoned	all 3 recovery attempts failed; last: validation still failing after 6s: health check failed
7:16:19 PM	A	needs_human_attention	records-api	3	abandoned	all 3 recovery attempts failed; last: validation still failing after 6s: health check failed
7:16:19 PM	C	needs_human_attention	records-api	3	abandoned	all 3 recovery attempts failed; last: validation still failing after 6s: health check failed
7:16:19 PM	B	needs_human_attention	records-api	3	abandoned	all 3 recovery attempts failed; last: validation still failing after 6s: health check failed
7:16:15 PM	D	RETRY_RECOVERY	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:16:14 PM	C	RETRY_RECOVERY	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:16:12 PM	B	RETRY_RECOVERY	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:16:11 PM	A	RETRY_RECOVERY	records-api	1	applied	the call succeeded and the cluster shows the effect
7:16:03 PM	D	RETRY_RECOVERY	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:16:01 PM	C	RETRY_RECOVERY	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:16:00 PM	B	RETRY_RECOVERY	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:58 PM	A	RETRY_RECOVERY	records-api	1	applied	the call succeeded and the cluster shows the effect
7:15:52 PM	D	CONTAIN/recover	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:52 PM	D	CONTAIN/isolate	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:50 PM	C	CONTAIN/recover	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:50 PM	C	CONTAIN/isolate	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:49 PM	B	CONTAIN/recover	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:49 PM	B	CONTAIN/isolate	records-api	1	already applied	already in place (this or another agent did it); nothing to do
7:15:47 PM	A	CONTAIN/recover	records-api	1	applied	the call succeeded and the cluster shows the effect
7:15:47 PM	A	CONTAIN/isolate	records-api	1	applied	the call succeeded and the cluster shows the effect

Pending votes (need 3 of 4 eligible signed votes; each agent says why it has not voted)
No proposals are waiting for votes.

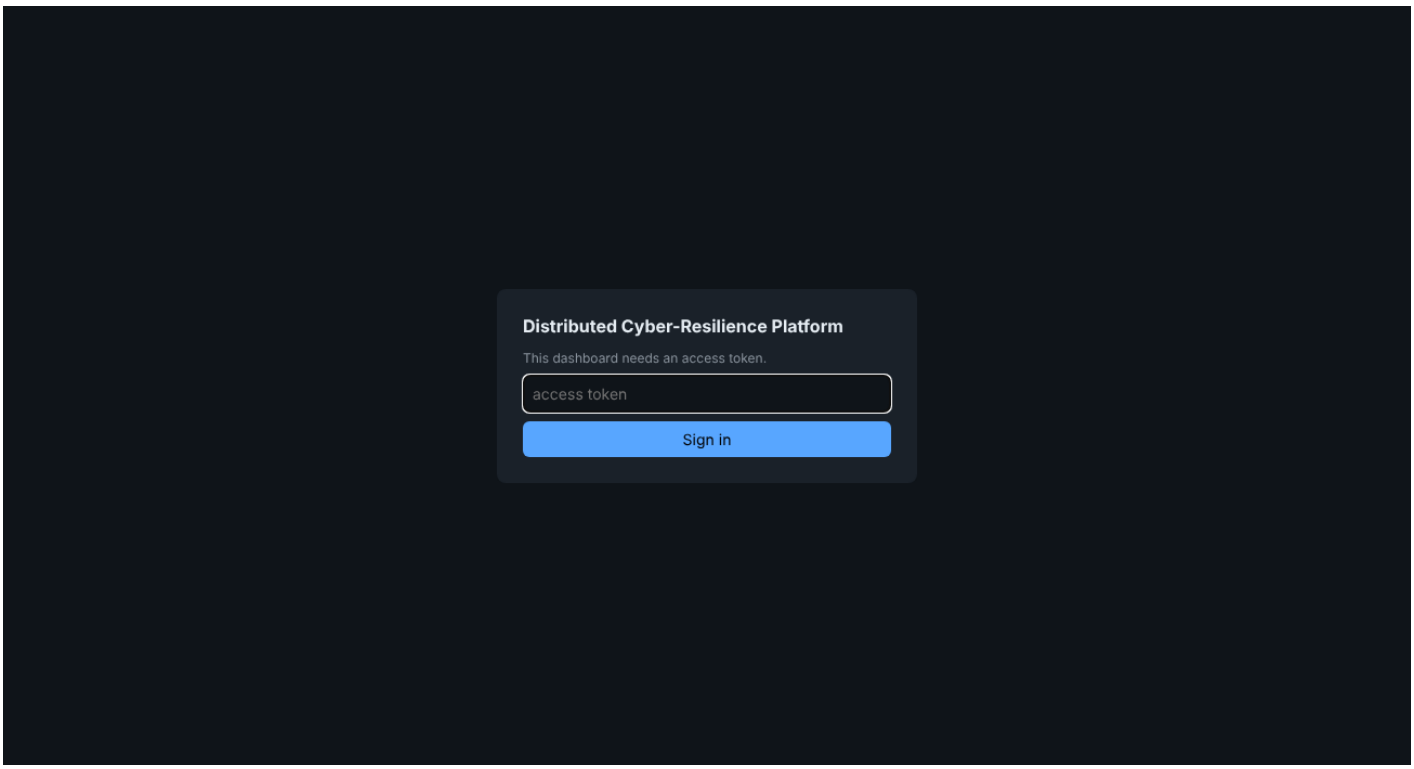
Rejected messages (signature / mTLS identity failures)

time	from (mTLS)	claimed	reason	seen by

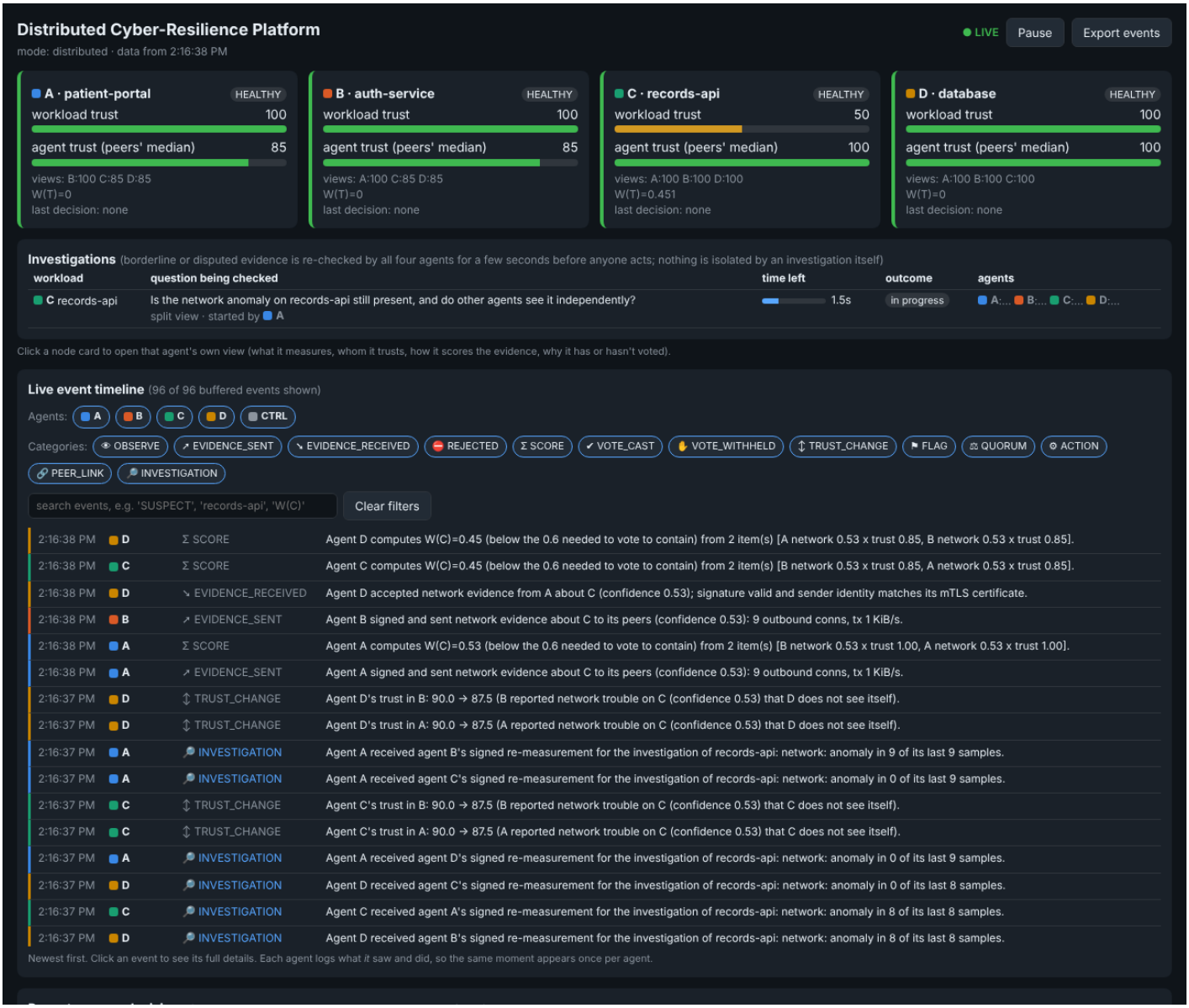
Incident metrics (seconds since injection; min across agents)

scenario	target	mode	TTD	TTI	TTR	validated	full	trust-recovery	flag attacker	false isolation	availability

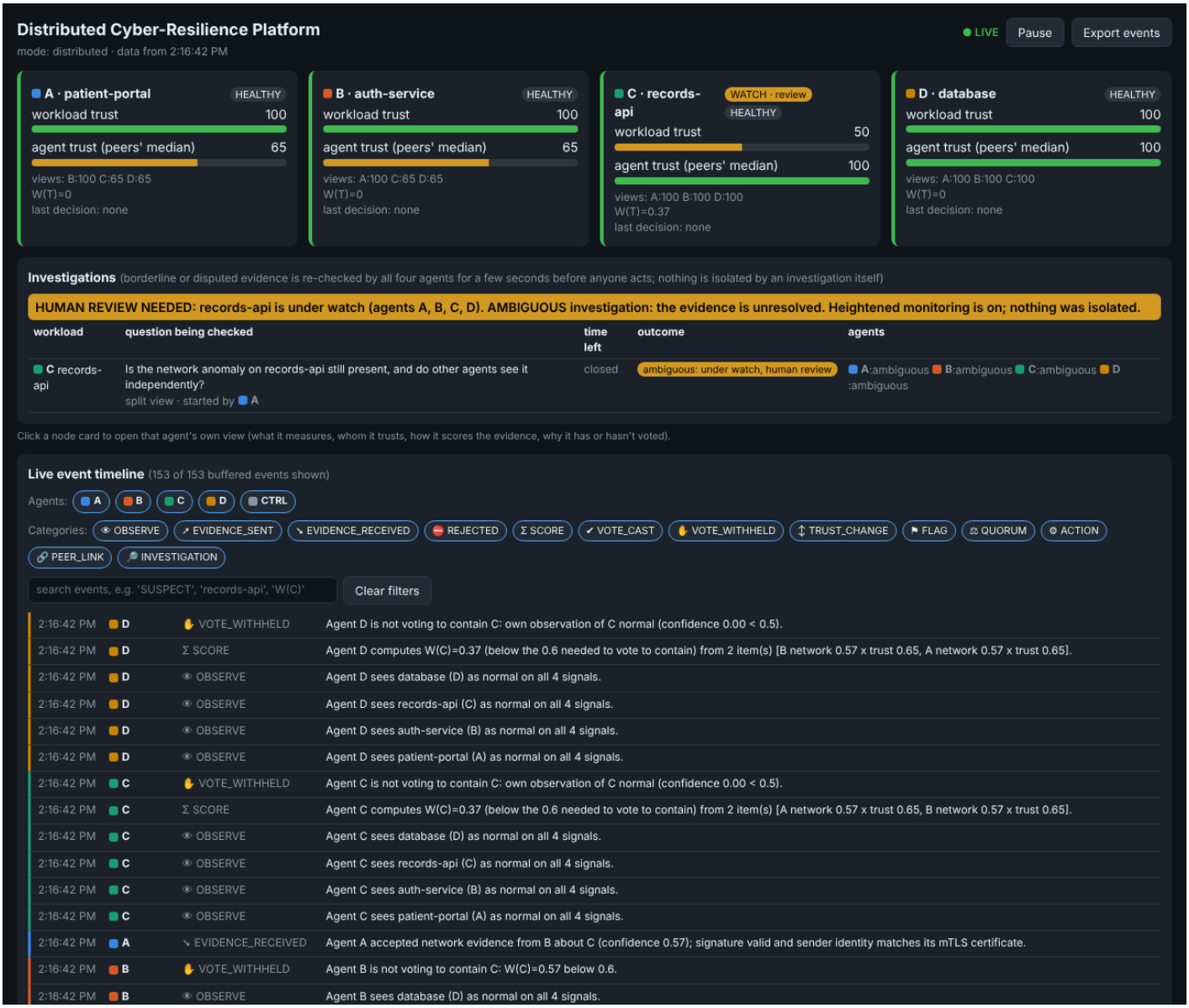
Early phases: dashboard access-token login



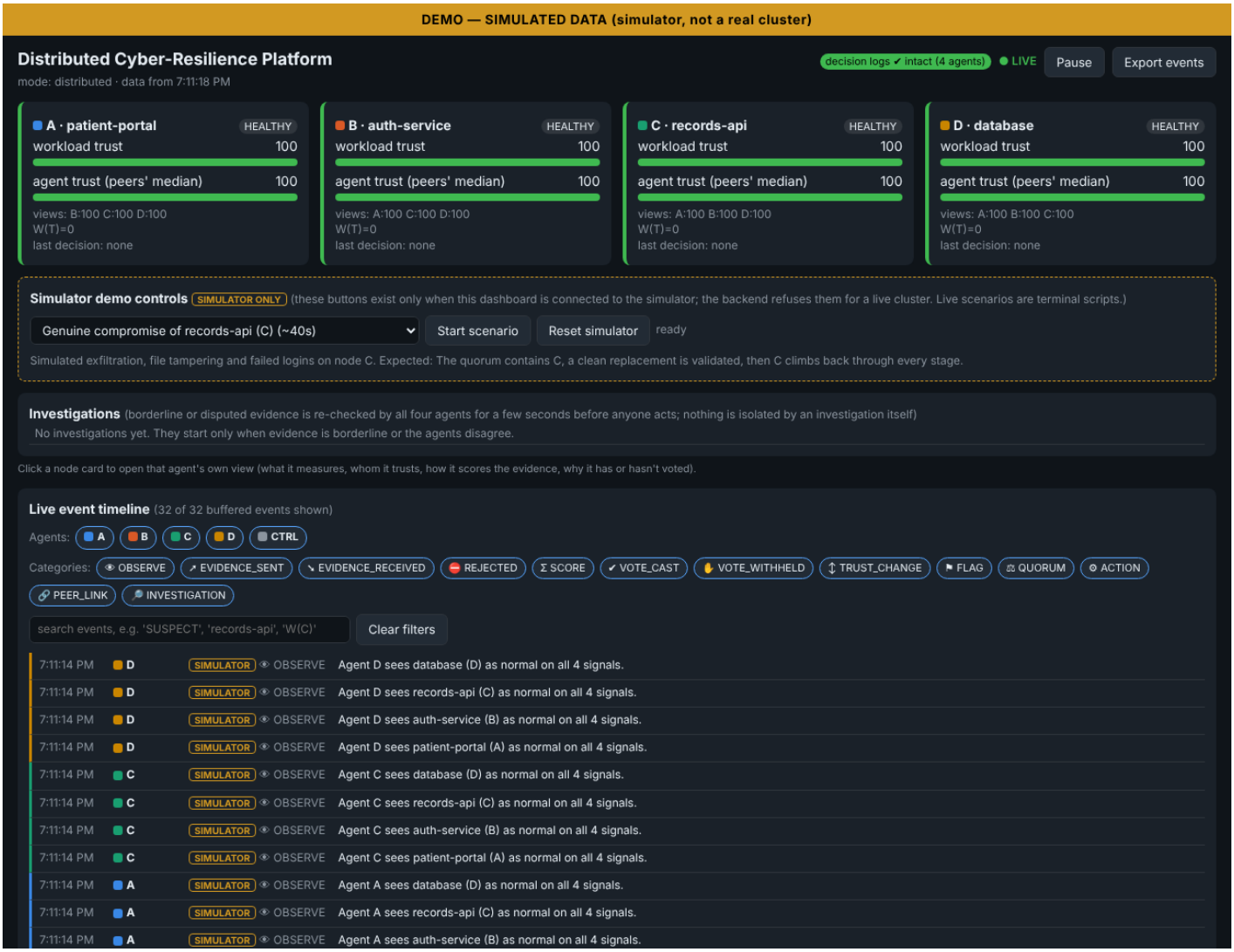
Phase 2: a targeted investigation in progress



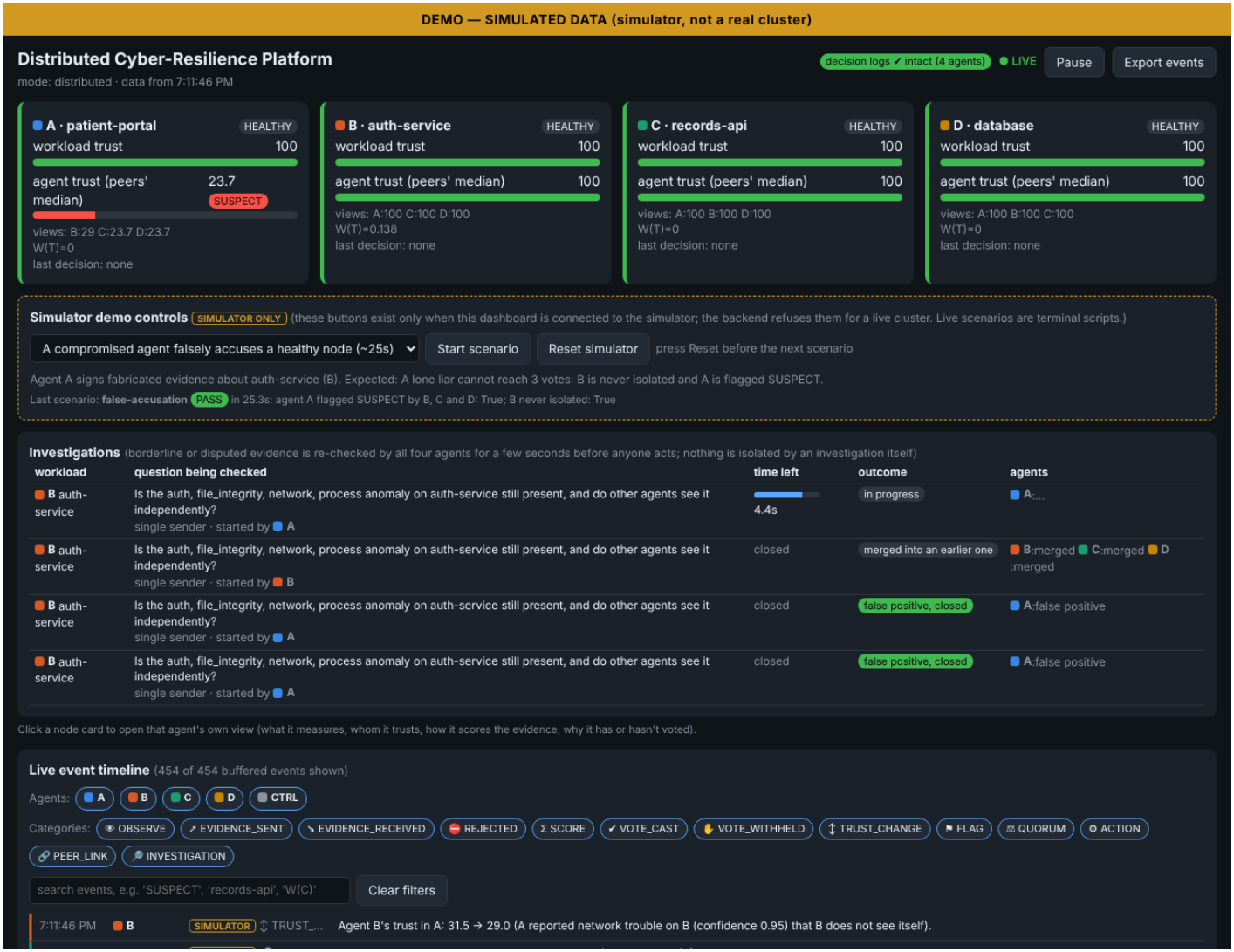
Phase 2: unresolved evidence - under watch, human review



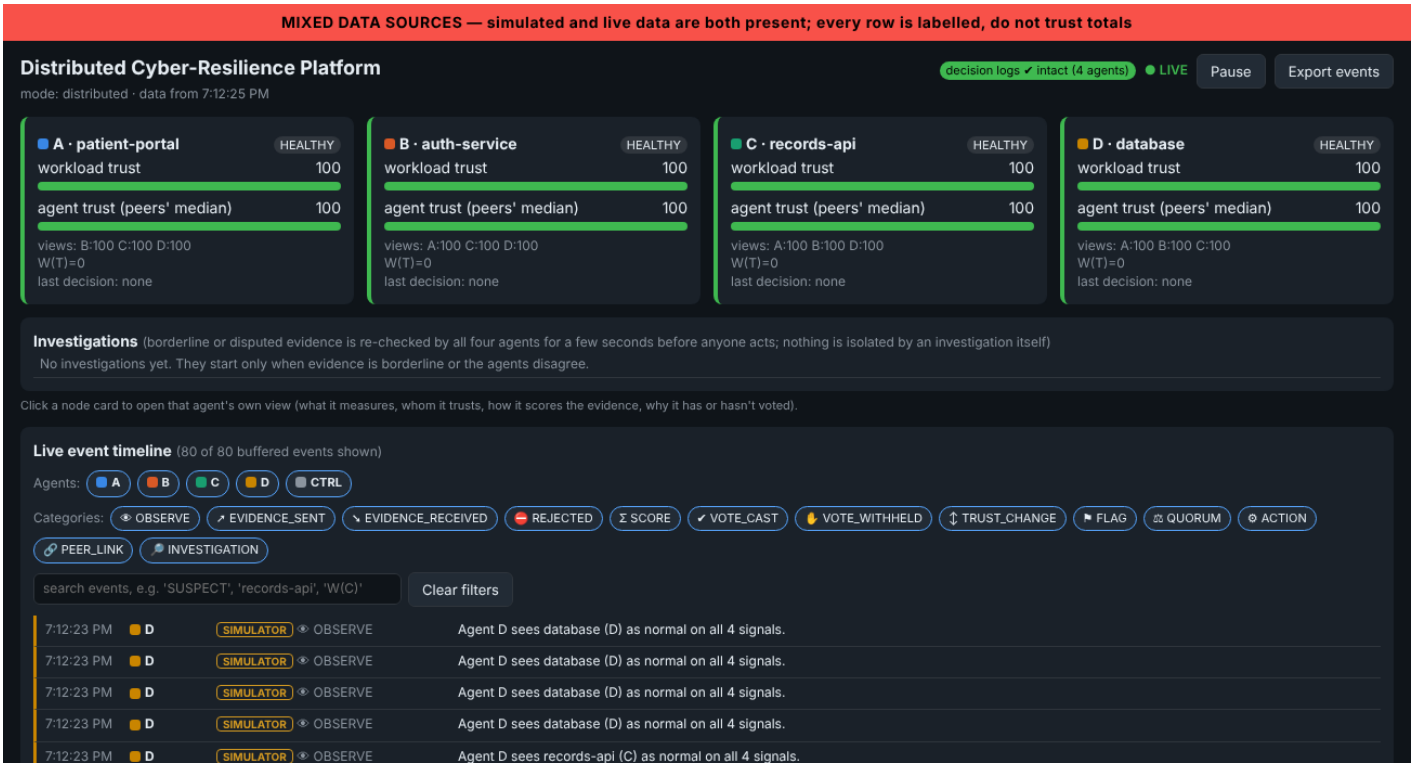
Dashboard Part 1: simulator demo controls (simulated data banner)



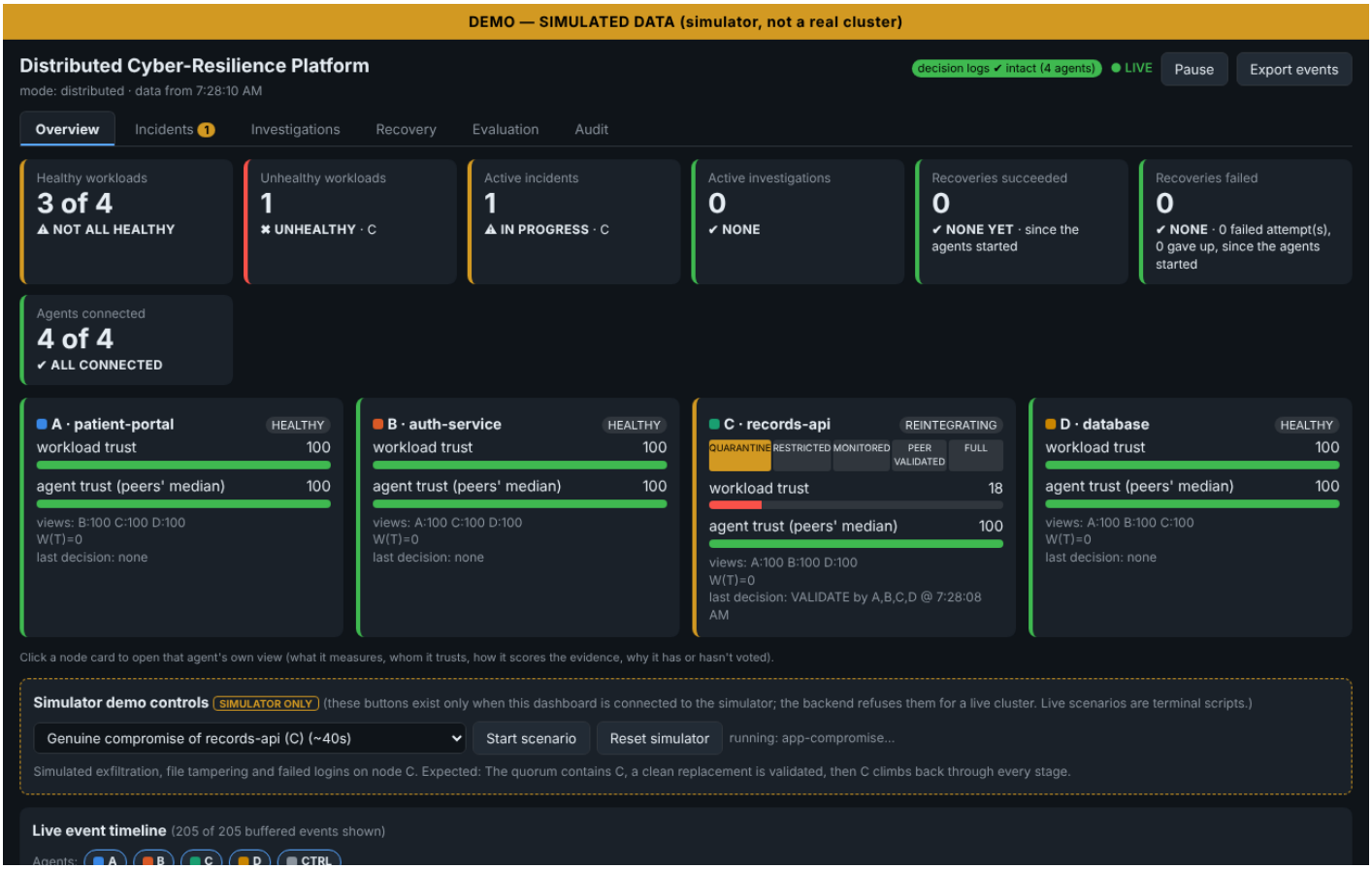
Dashboard Part 1: a scenario finished



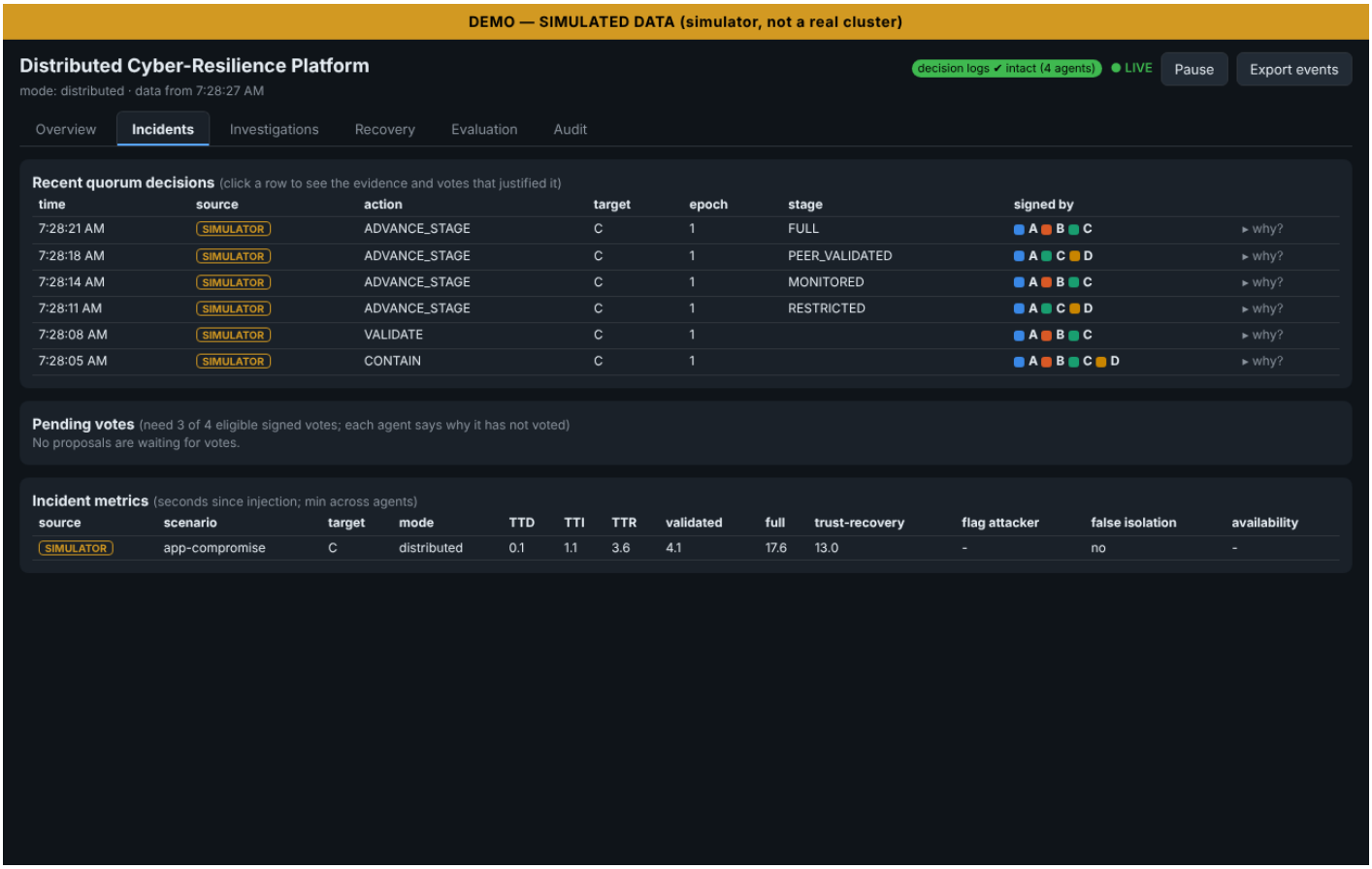
Dashboard Part 1: render check - mixed sources are flagged (test fixture)



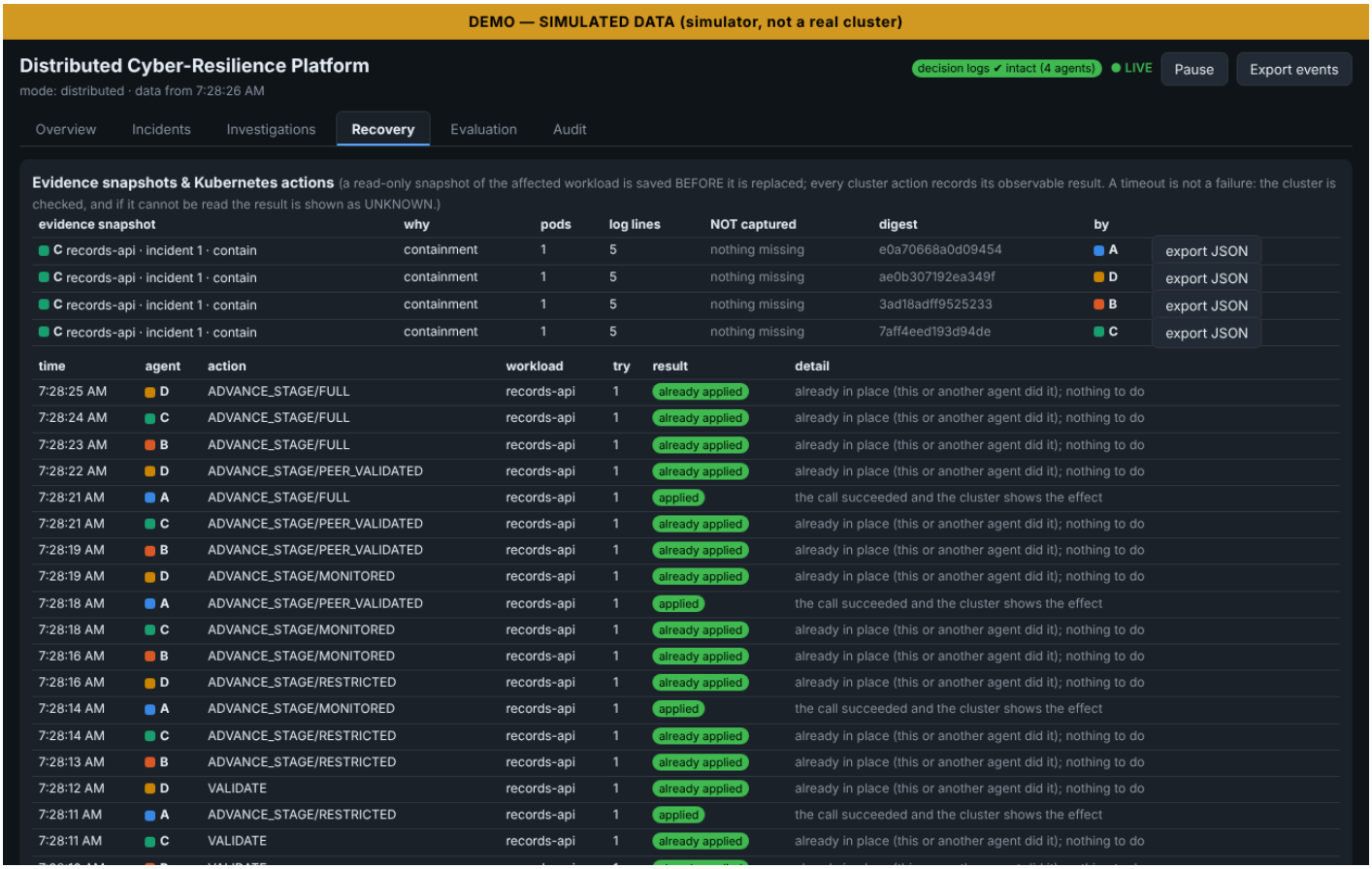
Dashboard Part 2: Overview with counters and an incident



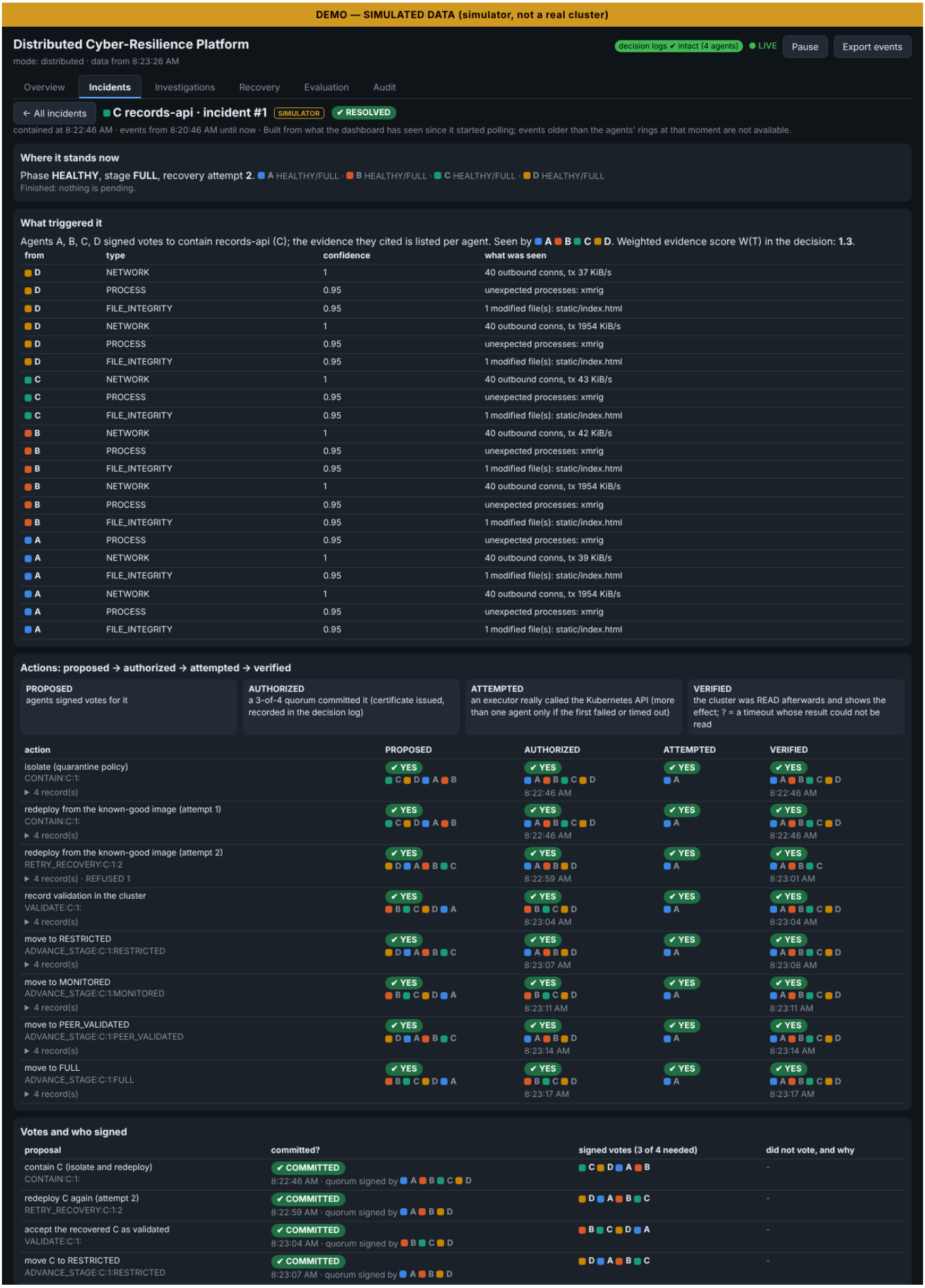
Dashboard Part 2: Incidents tab



Dashboard Part 2: Recovery tab



Dashboard Part 3: incident detail (validation-retry scenario) (part 1/2)



DEMO — SIMULATED DATA (simulator, not a real cluster)

Distributed Cyber-Resilience Platform

mode: distributed · data from 8:23:26 AM

decision logs ✔ intact (4 agents) · LIVE · Pause · Export events

Overview | Incidents | Investigations | Recovery | Evaluation | Audit

← All incidents · C records-api · incident #1 · SIMULATOR · ✔ RESOLVED

contained at 8:22:46 AM · events from 8:20:46 AM until now · Built from what the dashboard has seen since it started polling; events older than the agents' rings at that moment are not available.

Where it stands now

Phase **HEALTHY**, stage **FULL**, recovery attempt **2**. A HEALTHY/FULL · B HEALTHY/FULL · C HEALTHY/FULL · D HEALTHY/FULL

Finished: nothing is pending.

What triggered it

Agents A, B, C, D signed votes to contain records-api (C); the evidence they cited is listed per agent. Seen by A B C D. Weighted evidence score W(T) in the decision: **1.3**.

from	type	confidence	what was seen
D	NETWORK	1	40 outbound conns, tx 37 KiB/s
D	PROCESS	0.95	unexpected processes: xmrig
D	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
D	NETWORK	1	40 outbound conns, tx 1954 KiB/s
D	PROCESS	0.95	unexpected processes: xmrig
D	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
C	NETWORK	1	40 outbound conns, tx 43 KiB/s
C	PROCESS	0.95	unexpected processes: xmrig
C	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
B	NETWORK	1	40 outbound conns, tx 42 KiB/s
B	PROCESS	0.95	unexpected processes: xmrig
B	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
B	NETWORK	1	40 outbound conns, tx 1954 KiB/s
B	PROCESS	0.95	unexpected processes: xmrig
B	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
A	PROCESS	0.95	unexpected processes: xmrig
A	NETWORK	1	40 outbound conns, tx 39 KiB/s
A	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
A	NETWORK	1	40 outbound conns, tx 1954 KiB/s
A	PROCESS	0.95	unexpected processes: xmrig
A	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html

Actions: proposed → authorized → attempted → verified

PROPOSED agents signed votes for it

AUTHORIZED a 3-of-4 quorum committed it (certificate issued, recorded in the decision log)

ATTEMPTED an executor really called the Kubernetes API (more than one agent only if the first failed or timed out)

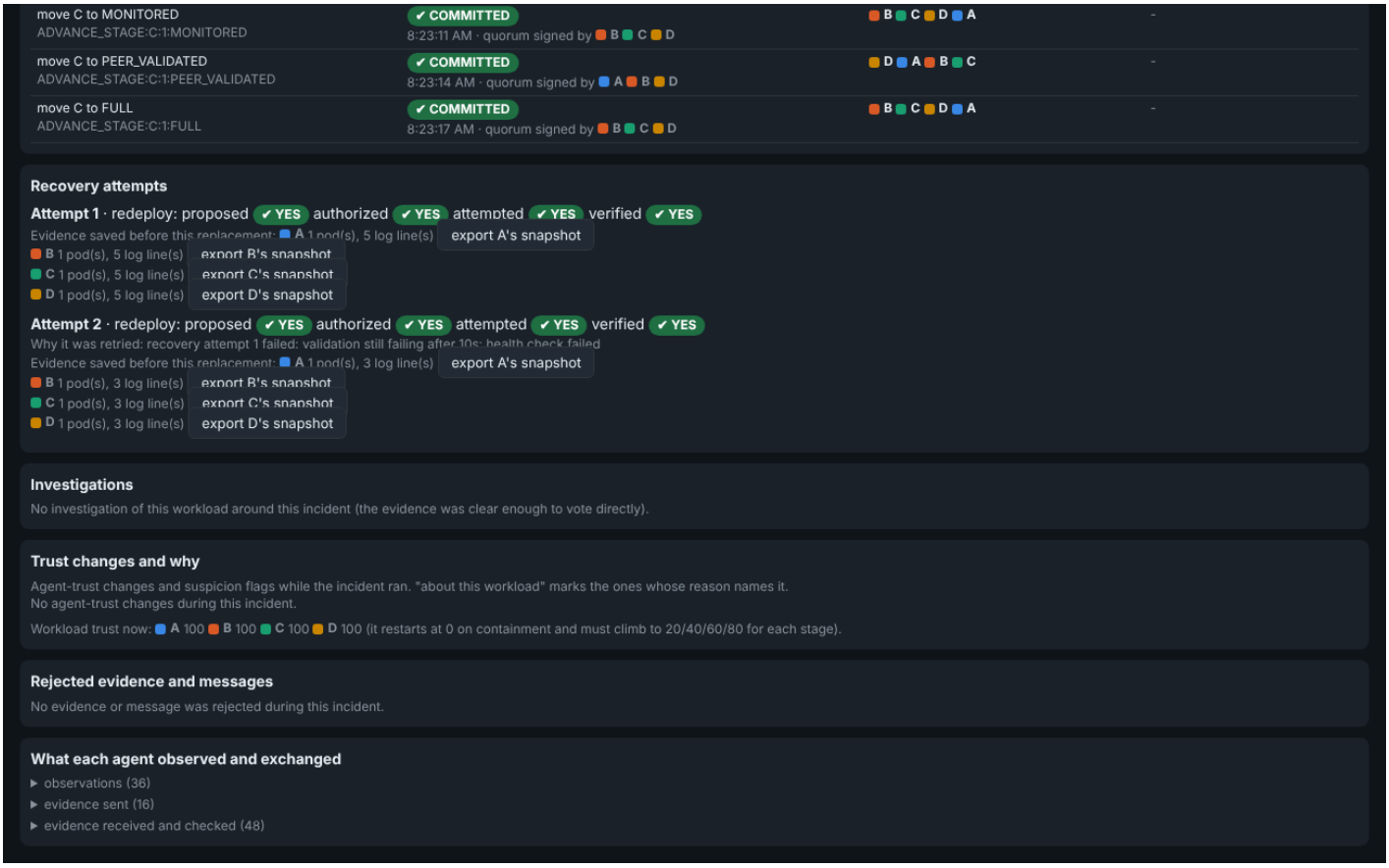
VERIFIED the cluster was READ afterwards and shows the effect; ? = a timeout whose result could not be read

action	PROPOSED	AUTHORIZED	ATTEMPTED	VERIFIED
isolate (quarantine policy) CONTAIN:C:1: ▶ 4 record(s)	✔ YES C D A B	✔ YES A B C D 8:22:46 AM	✔ YES A	✔ YES A B C D 8:22:46 AM
redeploy from the known-good image (attempt 1) CONTAIN:C:1: ▶ 4 record(s)	✔ YES C D A B	✔ YES A B C D 8:22:46 AM	✔ YES A	✔ YES A B C D 8:22:46 AM
redeploy from the known-good image (attempt 2) RETRY_RECOVERY:C:1:2 ▶ 4 record(s) · REFUSED 1	✔ YES D A B C	✔ YES A B D 8:22:59 AM	✔ YES A	✔ YES A B C 8:23:01 AM
record validation in the cluster VALIDATE:C:1: ▶ 4 record(s)	✔ YES B C D A	✔ YES B C D 8:23:04 AM	✔ YES A	✔ YES A B C D 8:23:04 AM
move to RESTRICTED ADVANCE_STAGE:C:1:RESTRICTED ▶ 4 record(s)	✔ YES D A B C	✔ YES A B D 8:23:07 AM	✔ YES A	✔ YES A B C D 8:23:08 AM
move to MONITORED ADVANCE_STAGE:C:1:MONITORED ▶ 4 record(s)	✔ YES B C D A	✔ YES B C D 8:23:11 AM	✔ YES A	✔ YES A B C D 8:23:11 AM
move to PEER_VALIDATED ADVANCE_STAGE:C:1:PEER_VALIDATED ▶ 4 record(s)	✔ YES D A B C	✔ YES A B D 8:23:14 AM	✔ YES A	✔ YES A B C D 8:23:14 AM
move to FULL ADVANCE_STAGE:C:1:FULL ▶ 4 record(s)	✔ YES B C D A	✔ YES B C D 8:23:17 AM	✔ YES A	✔ YES A B C D 8:23:17 AM

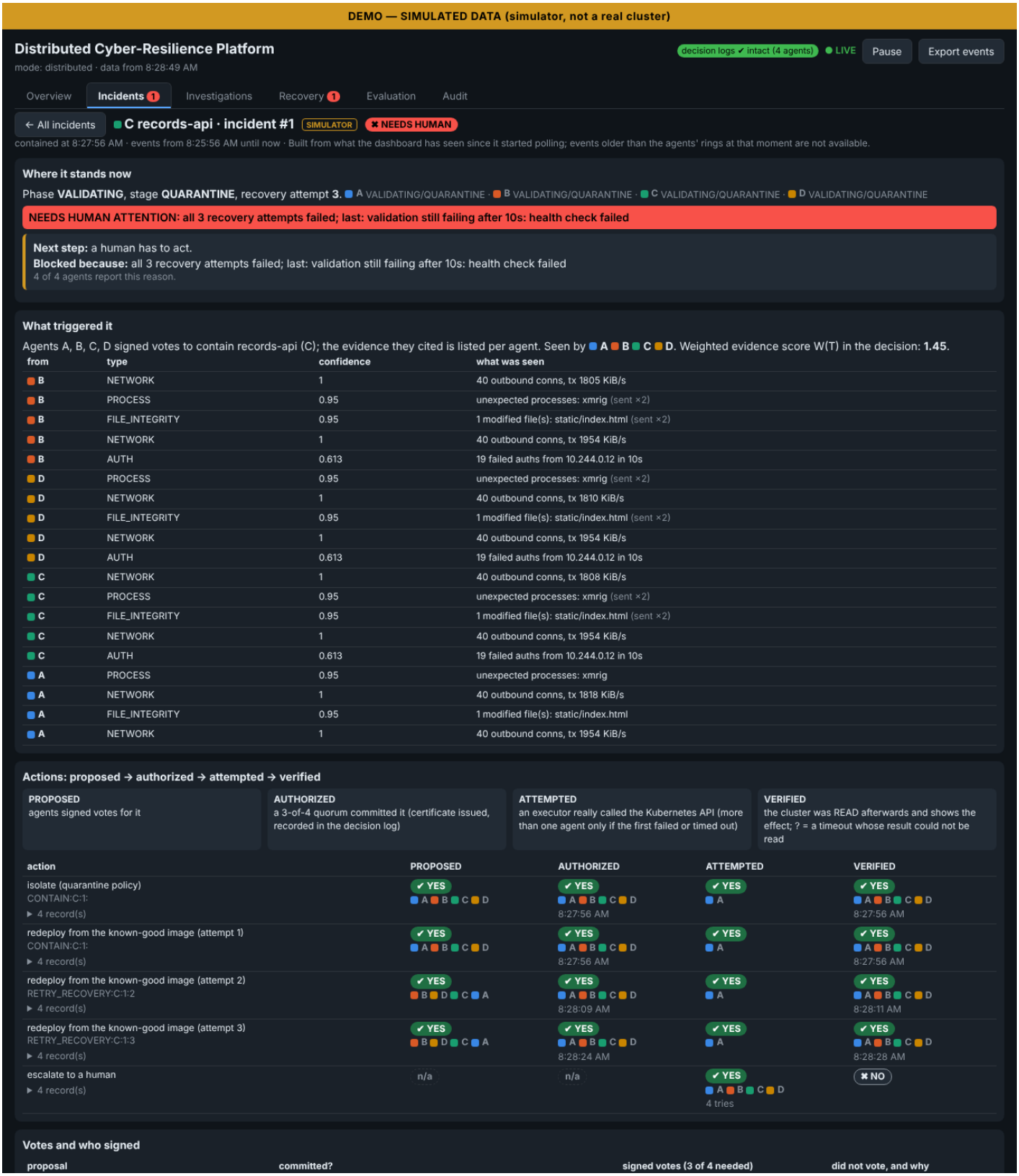
Votes and who signed

proposal	committed?	signed votes (3 of 4 needed)	did not vote, and why
contain C (isolate and redeploy) CONTAIN:C:1:	✔ COMMITTED 8:22:46 AM · quorum signed by A B C D	C D A B	-
redeploy C again (attempt 2) RETRY_RECOVERY:C:1:2	✔ COMMITTED 8:22:59 AM · quorum signed by A B D	D A B C	-
accept the recovered C as validated VALIDATE:C:1:	✔ COMMITTED 8:23:04 AM · quorum signed by B C D	B C D A	-
move C to RESTRICTED ADVANCE_STAGE:C:1:RESTRICTED	✔ COMMITTED 8:23:07 AM · quorum signed by A B D	D A B C	-

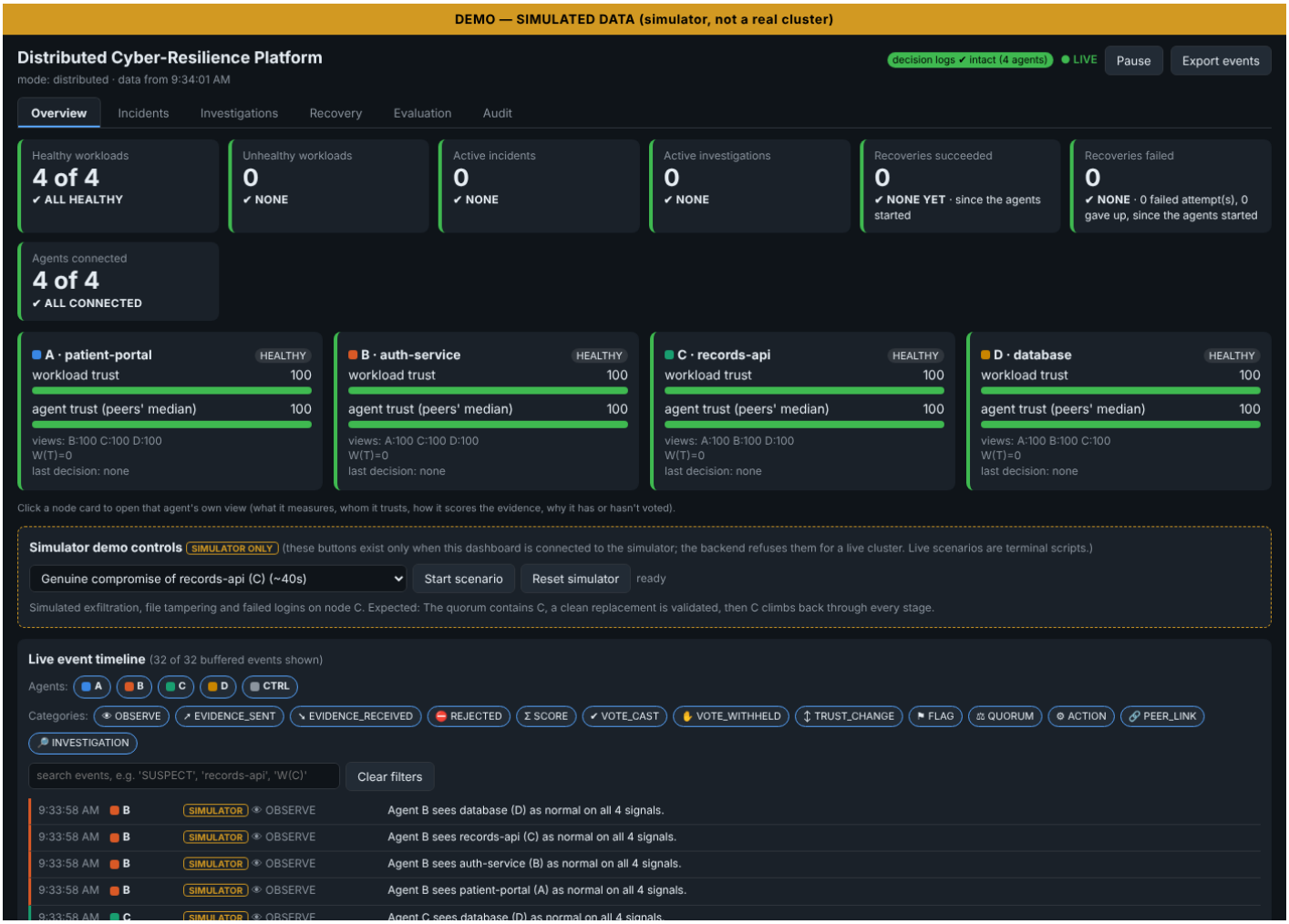
Dashboard Part 3: incident detail (validation-retry scenario) (part 2/2)



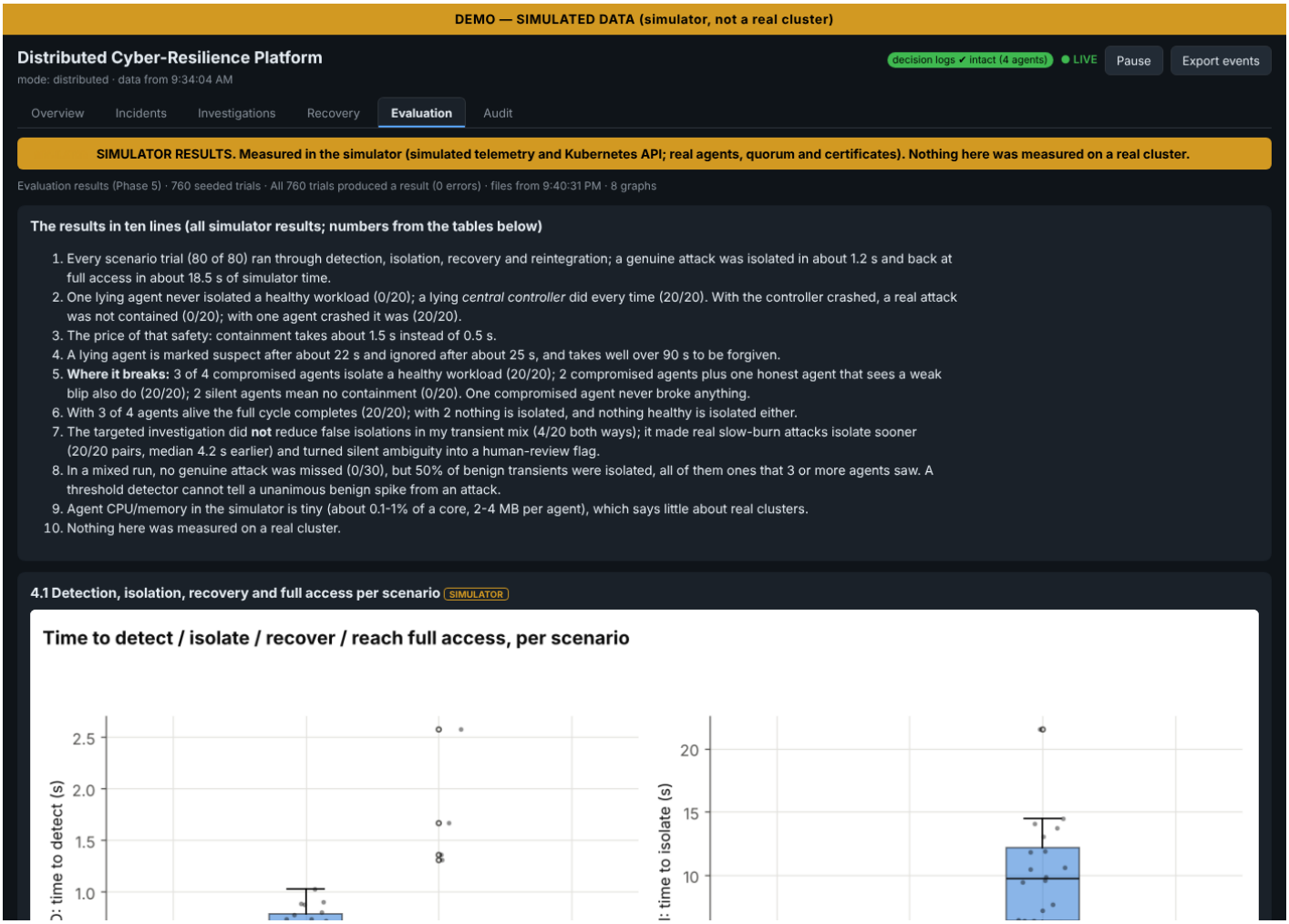
Dashboard Part 3: incident that needs human attention



Dashboard Part 4: Overview



Dashboard Part 4: Evaluation tab (simulator results)



Dashboard Part 4: Evaluation tab, full page (before Phase C/D changes) (part 1/6)

DEMO — SIMULATED DATA (simulator, not a real cluster)

Distributed Cyber-Resilience Platform

mode: distributed · data from 9:34:04 AM

decision logs ✔ intact (4 agents) · LIVE · Pause · Export events

Overview | Incidents | Investigations | Recovery | **Evaluation** | Audit

SIMULATOR RESULTS. Measured in the simulator (simulated telemetry and Kubernetes API; real agents, quorum and certificates). Nothing here was measured on a real cluster.

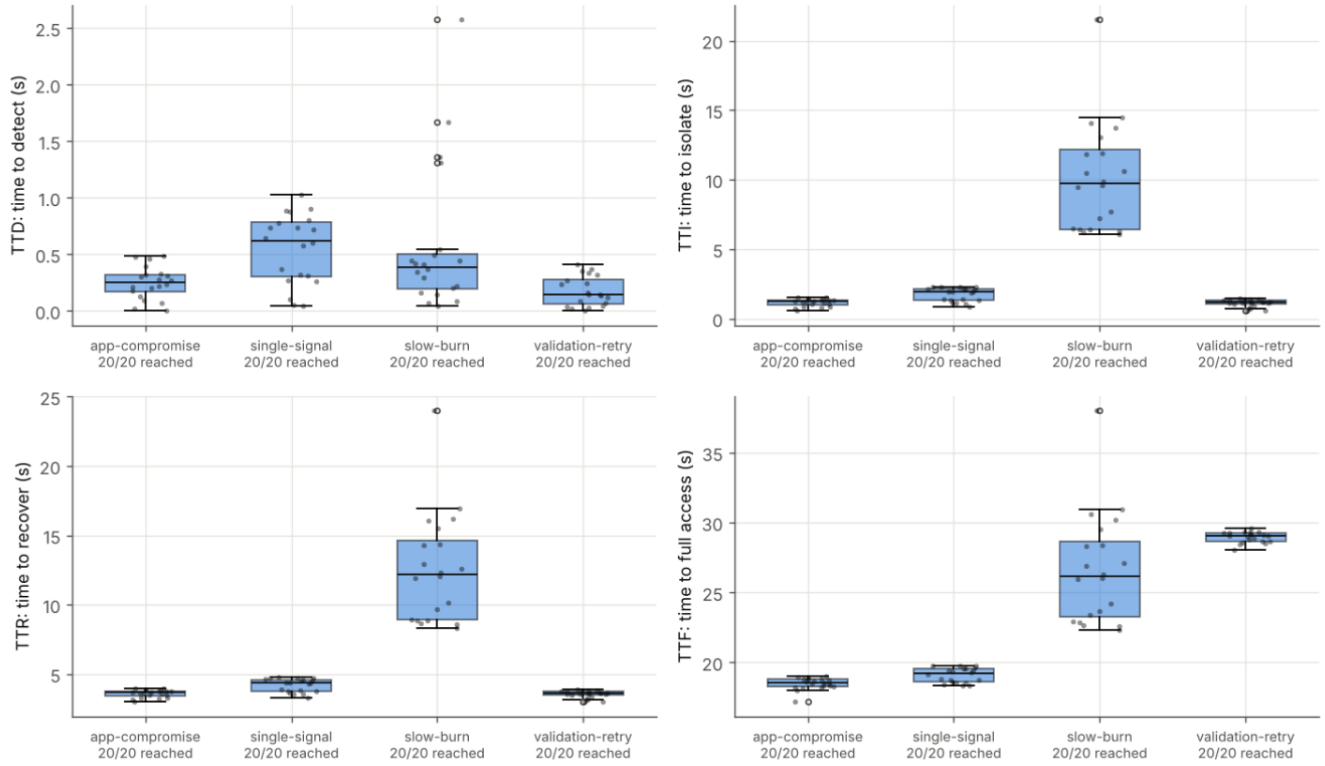
Evaluation results (Phase 5) · 760 seeded trials · All 760 trials produced a result (0 errors) · files from 9:40:31 PM · 8 graphs

The results in ten lines (all simulator results; numbers from the tables below)

1. Every scenario trial (80 of 80) ran through detection, isolation, recovery and reintegration; a genuine attack was isolated in about 1.2 s and back at full access in about 18.5 s of simulator time.
2. One lying agent never isolated a healthy workload (0/20); a lying *central controller* did every time (20/20). With the controller crashed, a real attack was not contained (0/20); with one agent crashed it was (20/20).
3. The price of that safety: containment takes about 1.5 s instead of 0.5 s.
4. A lying agent is marked suspect after about 22 s and ignored after about 25 s, and takes well over 90 s to be forgiven.
5. **Where it breaks:** 3 of 4 compromised agents isolate a healthy workload (20/20); 2 compromised agents plus one honest agent that sees a weak blip also do (20/20); 2 silent agents mean no containment (0/20). One compromised agent never broke anything.
6. With 3 of 4 agents alive the full cycle completes (20/20); with 2 nothing is isolated, and nothing healthy is isolated either.
7. The targeted investigation did **not** reduce false isolations in my transient mix (4/20 both ways); it made real slow-burn attacks isolate sooner (20/20 pairs, median 4.2 s earlier) and turned silent ambiguity into a human-review flag.
8. In a mixed run, no genuine attack was missed (0/30), but 50% of benign transients were isolated, all of them ones that 3 or more agents saw. A threshold detector cannot tell a unanimous benign spike from an attack.
9. Agent CPU/memory in the simulator is tiny (about 0.1-1% of a core, 2-4 MB per agent), which says little about real clusters.
10. Nothing here was measured on a real cluster.

4.1 Detection, isolation, recovery and full access per scenario SIMULATOR

Time to detect / isolate / recover / reach full access, per scenario



Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Scenario	Trials	Reached full access	TTD (s)	TTI (s)	TTR (s)	TTF (s)
app-compromise	20	20/20	median 0.3 (IQR 0.2-0.3 min 0.0, max 0.5, n=20)	median 1.2 (IQR 1.0-1.4 min 0.6, max 1.6, n=20)	median 3.7 (IQR 3.4-3.8 min 3.0, max 4.0, n=20)	median 18.5 (IQR 18.3-18.8 min 17.2, max 19.0, n=20)
single-signal	20	20/20	median 0.6 (IQR 0.3-0.8 min 0.0, max 1.0, n=20)	median 1.9 (IQR 1.3-2.1 min 0.9, max 2.3, n=20)	median 4.4 (IQR 3.8-4.6 min 3.3, max 4.8, n=20)	median 19.2 (IQR 18.6-19.5 min 18.3, max 19.7, n=20)
slow-burn	20	20/20	median 0.4 (IQR 0.2-0.5 min 0.0, max 2.6, n=20)	median 9.7 (IQR 6.4-12.2 min 6.1, max 21.5, n=20)	median 12.2 (IQR 8.9-14.7 min 8.3, max 24.0, n=20)	median 26.2 (IQR 23.3-28.7 min 22.3, max 38.0, n=20)
validation-retry	20	20/20	median 0.1 (IQR 0.1-0.3 min 0.0, max 0.4, n=20)	median 1.2 (IQR 1.0-1.3 min 0.6, max 1.5, n=20)	median 3.6 (IQR 3.5-3.7 min 3.0, max 3.9, n=20)	median 29.1 (IQR 28.7-29.3 min 28.0, max 29.6, n=20)

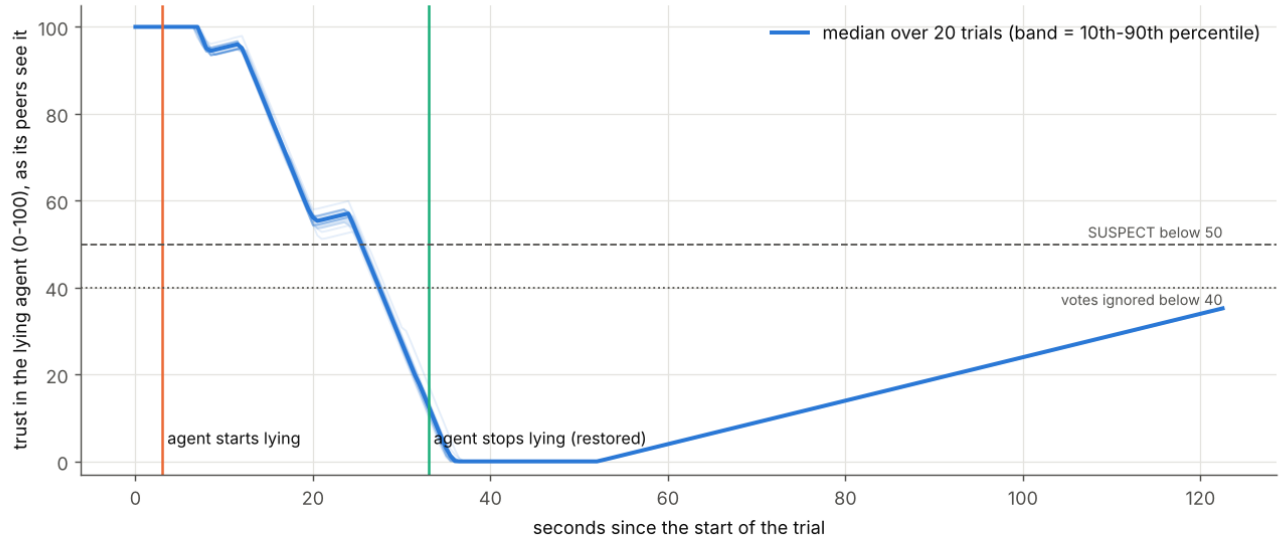
How to read it. Each box is one scenario over its trials: the line is the median, the box covers the middle half (25th-75th percentile), the whiskers the rest apart from outliers (dots). Under each name is "x/y reached": how many trials got that far. Lower is faster. Times are small because the simulator compresses all timers (0.5 s ticks, 3 s per reintegration stage, 2 s for a "pod" to start); relative differences between scenarios, not the seconds, are the point. **What the data shows.** All 80 trials reached full access (no trial stalled). A genuine multi-signal compromise was detected in a median 0.3 s, isolated in 1.2 s, recovered in 3.7 s and back at full access in 18.5 s (most of the time after recovery is the four staged reintegration steps, each with a 3 s minimum dwell, plus the trust-gated waits). A single-signal attack is slightly slower to isolate (1.9 s): I did not isolate the cause; one weaker signal plausibly takes a little longer to gather three agreeing votes. The **slow burn** is the slowest to isolate (median

Dashboard Part 4: Evaluation tab, full page (before Phase C/D changes) (part 2/6)

9.7 s, up to 21.5 s) because the signal starts weak and only some agents see it first; the platform waits for it to persist and grow, which is the intended trade-off. **Validation-retry** isolates and recovers as fast as a normal attack (its TTR is the *first*, bad, replacement) but reaches full access later (29.1 s against 18.5 s): one failed validation plus back-off costs about 10 s in simulator time. Timing spreads are narrow because the simulator is largely deterministic; real systems will be noisier.

4.2 Trust in a lying agent SIMULATOR

Trust in a lying agent: decay while it lies, slow recovery after



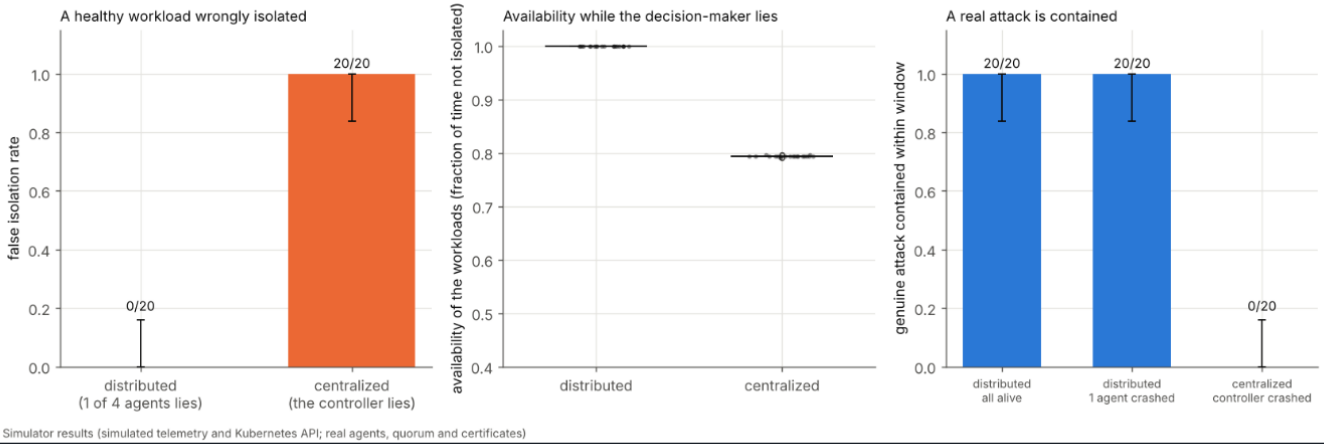
Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Quantity	Result
Trials	20
Time until the first honest agent marks the liar SUSPECT (<50), s after it starts lying	median 22.5 (IQR 22.5-22.5, min 17.5, max 23.0, n=20)
Time until a vote-exclusion (<40)	median 24.5 (IQR 24.5-24.5, min 24.0, max 25.0, n=20)
Lowest trust any honest agent reached	median 0.0 (IQR 0.0-0.0, min 0.0, max 0.0, n=20)
Time after it stops lying until mean trust is back above 50, s	n/a (no trial reached it)
Mean trust at the end (90 s after it stopped)	median 35.3 (IQR 35.2-35.3, min 35.0, max 35.3, n=20)
Healthy victim wrongly isolated	0/20 = 0% (95% CI 0-16%)

How to read it. One thin line per trial is the average trust the three honest agents place in the liar. The thick line is the median, the band the 10th-90th percentile. Trust starts at 100. While the agent lies, its claims are contradicted by what the honest agents see themselves, so its trust falls. It crosses the SUSPECT line (50) and then the vote-exclusion line (40), after which its votes and evidence are ignored. When it stops lying, trust climbs back, but slowly (0.5 per second): the platform is deliberately slow to forgive. **What the data shows.** The honest agents first marked the liar SUSPECT about 22.5 s after it started lying (range 17.5-23 s) and stopped counting its votes about 2 s later (24.5 s). Trust fell to 0 in every trial. In **no** trial did the lie isolate the victim (0/20, 95% CI 0-16%), because one liar never reaches 3 signatures. After the agent was restored, trust stayed at 0 for about 15 s (its old fabricated evidence was still inside the 20 s evidence window), then climbed at 0.5 per second and was only about 35 after 90 s; **no trial got back above 50 within the observation**. Going from 0 to 50 takes at least 100 s at that rate. In other words forgiveness is deliberately much slower than suspicion. Note the 20 curves nearly coincide because the simulated dynamics are deterministic; the spread in a real cluster would be wider. Also note the first ~20 s: the platform is robust to the lie *because of the quorum*, not because trust has already dropped.

4.3 Distributed vs centralized SIMULATOR

Distributed (3-of-4 quorum) vs centralized controller



Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Condition	Result
Distributed, 1 agent lies about a healthy workload: false isolation	0/20 = 0% (95% CI 0-16%)

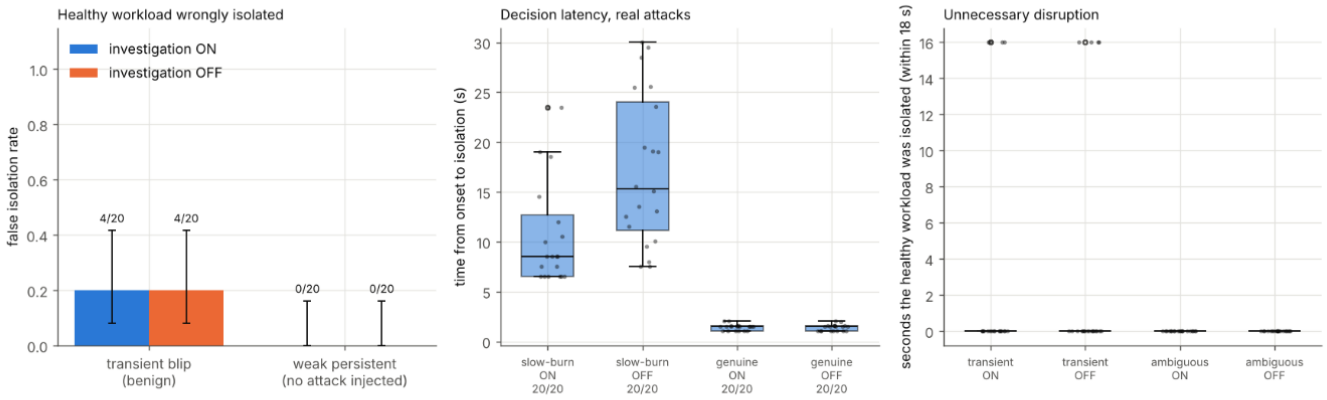
Dashboard Part 4: Evaluation tab, full page (before Phase C/D changes) (part 3/6)

Centralized, the controller lies: false isolation	20/20 = 100% (95% CI 84-100%)
Distributed, availability while an agent lies	median 1.000 (IQR 1.000-1.000, min 1.000, max 1.000, n=20)
Centralized, availability while the controller lies	median 0.794 (IQR 0.794-0.794, min 0.791, max 0.797, n=20)
Distributed, all alive: real attack contained	20/20 = 100% (95% CI 84-100%)
Distributed, 1 agent crashed first: real attack contained	20/20 = 100% (95% CI 84-100%)
Centralized, controller alive: real attack contained	20/20 = 100% (95% CI 84-100%)
Centralized, controller crashed first: real attack contained	0/20 = 0% (95% CI 0-16%)
Distributed, time to contain a real attack (s)	median 1.50 (IQR 1.25-1.51, min 1.01, max 1.75, n=20)
Centralized, time to contain a real attack (s)	median 0.50 (IQR 0.50-0.50, min 0.50, max 0.76, n=20)

How to read it. Left: when the single decision-maker is compromised and lies, how often is a healthy workload wrongly isolated? Distributed: one of four agents lies (it cannot reach a quorum alone). Centralized: the one controller lies (its word is final). Middle: the fraction of time the workloads stayed un-isolated while this went on (higher is better; only the lie, not a real attack, is happening). Right: when a real attack occurs, how often is it contained, with all components alive, with one agent crashed, and with the single controller crashed. Whiskers are 95% confidence intervals.
What the data shows. When the single decision-maker lies, the centralized controller isolated a healthy workload in **20 of 20** trials (95% CI 84-100%), whereas with one lying agent among four the distributed platform did so in **0 of 20** (CI 0-16%). While it lied, the centralized system's workloads were available only 79% of the time (the accused workload was cut off for about 33 of the 40 s window); the distributed platform kept 100%. When the decision-maker **crashes** before a real attack, the centralized controller contained it in **0 of 20** trials, whereas the distributed platform with one agent crashed still contained **20 of 20**. The price: with all components healthy the centralized controller contained a real attack faster (median 0.5 s against 1.5 s), because a single controller needs no signatures, votes or certificate; the distributed platform spends about one extra second to be safe against a lying or crashed member. (The centralized baseline deliberately has no quorum or certificates; it is the comparison, not a design I am defending.)

4.4 Targeted investigation ON vs OFF SIMULATOR

Targeted investigation ON vs OFF on identical inputs (same seeds)



Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Input	Investigation	Trials	Workload isolated	Isolated seconds (median, of 18 s)	Onset-to-isolation latency (s)	Human-review flag
transient (benign, no attack)	ON	20	4/20 = 20% (95% CI 8-42%)	0.0	n/a (benign input)	0/20
transient (benign, no attack)	OFF	20	4/20 = 20% (95% CI 8-42%)	0.0	n/a (benign input)	0/20
ambiguous (benign, no attack)	ON	20	0/20 = 0% (95% CI 0-16%)	0.0	n/a (benign input)	20/20
ambiguous (benign, no attack)	OFF	20	0/20 = 0% (95% CI 0-16%)	0.0	n/a (benign input)	0/20
slow-burn (real attack)	ON	20	20/20 = 100% (95% CI 84-100%)	n/a	median 8.5 (IQR 6.5-12.7, min 6.5, max 23.5, n=20)	0/20
slow-burn (real attack)	OFF	20	20/20 = 100% (95% CI 84-100%)	n/a	median 15.3 (IQR 11.1-24.1, min 7.5, max 30.0, n=20)	0/20
genuine (real attack)	ON	20	20/20 = 100% (95% CI 84-100%)	n/a	median 1.5 (IQR 1.1-1.5, min 1.0, max 2.0, n=20)	0/20
genuine (real attack)	OFF	20	20/20 = 100% (95% CI 84-100%)	n/a	median 1.5 (IQR 1.1-1.6, min 1.0, max 2.0, n=20)	0/20

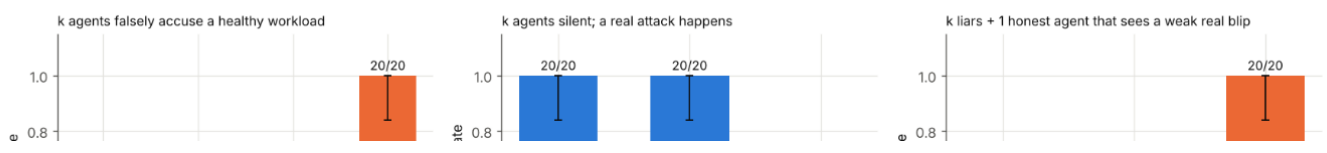
How to read it. Each pair of bars/boxes uses **identical inputs** (same seeds); only the investigation switch differs. Left: for benign inputs, how often was a healthy workload isolated. Middle: for real attacks, how long from onset to isolation (does checking first cost time?). Right: for benign inputs, how many seconds the healthy workload was cut off. The "weak persistent" input is a signal that no attack caused and that only two agents confirm; with the investigation ON it ends in a human-review flag instead of an action. **What the data shows, including what it does not show.**

- **Transient blip (benign):** 4 of 20 healthy workloads were isolated with the investigation ON and the same 4 with it OFF (identical trials; each was a spike seen by 3 of 4 agents, which satisfies the quorum on its own). Where only 1-2 agents saw the blip, neither configuration isolated anything, because the 3-vote quorum alone already refuses. So **in this input mix the investigation did not reduce false isolations**; it is not the thing that stops a 1-2-agent blip, the quorum is. I did not measure a benefit here and do not claim one.
- **Weak persistent signal (no attack injected, confirmed by 2 agents):** no isolation in either mode (0/20 each), but with the investigation ON all 20 ended in an explicit **human-review flag**, and with it OFF nothing was surfaced at all. The measurable gain is visibility, not fewer actions.
- **Slow burn (real attack):** with the investigation ON isolation came earlier in **20 of 20** paired trials (median 8.5 s against 15.3 s; the per-seed difference has median 4.2 s, range 0.5-19.0 s). Its targeted re-measurement lets agents that see a persistent weak anomaly vote sooner.
- **Genuine strong attack:** identical (median 1.5 s both): the investigation adds no delay when evidence is already clear.

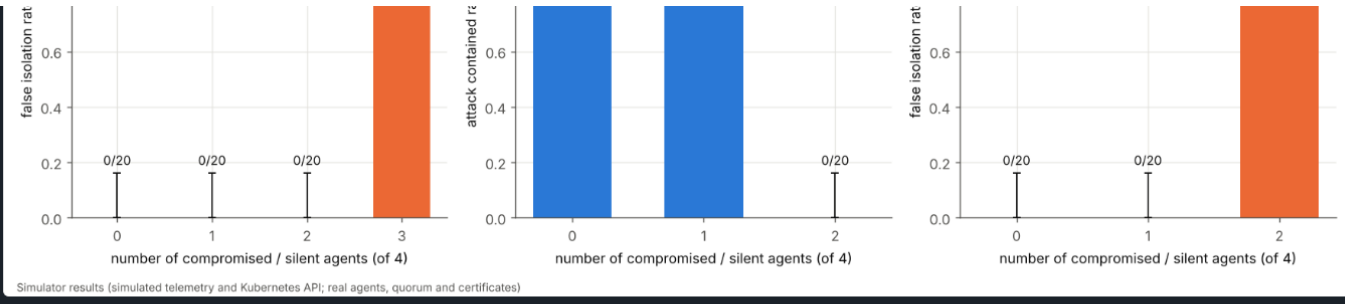
Unnecessary disruption: the 4 isolated transient trials were each isolated for the rest of the 18 s window (right-censored; the dots on the right graph).

4.5 Fault-tolerance boundary SIMULATOR

Fault-tolerance boundary: 0, 1, 2 (and 3) of 4 agents compromised or silent



Dashboard Part 4: Evaluation tab, full page (before Phase C/D changes) (part 4/6)



Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Test	Compromised / silent agents (k)	Trials	Outcome
k liars accuse a healthy workload: **false isolation**	0	20	0/20 = 0% (95% CI 0-16%)
k liars accuse a healthy workload: **false isolation**	1	20	0/20 = 0% (95% CI 0-16%)
k liars accuse a healthy workload: **false isolation**	2	20	0/20 = 0% (95% CI 0-16%)
k liars accuse a healthy workload: **false isolation**	3	20	20/20 = 100% (95% CI 84-100%)
k liars + 1 honest agent that sees a weak real blip: **false isolation**	0	20	0/20 = 0% (95% CI 0-16%)
k liars + 1 honest agent that sees a weak real blip: **false isolation**	1	20	0/20 = 0% (95% CI 0-16%)
k liars + 1 honest agent that sees a weak real blip: **false isolation**	2	20	20/20 = 100% (95% CI 84-100%)
k agents silent, real attack: **contained**	0	20	20/20 = 100% (95% CI 84-100%)
k agents silent, real attack: **contained**	1	20	20/20 = 100% (95% CI 84-100%)
k agents silent, real attack: **contained**	2	20	0/20 = 0% (95% CI 0-16%)

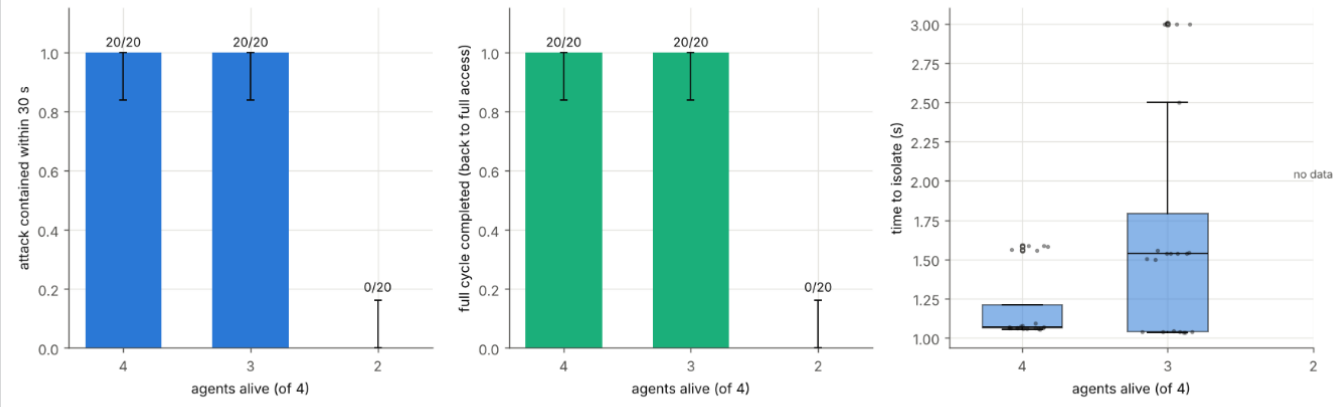
How to read it. Byzantine fault-tolerance theory says that with 4 agents and a quorum of 3, **1** compromised agent can be tolerated and **2** cannot (n >= 3f+1). Three tests probe the two ways it can fail. *Left (safety):* k compromised agents all falsely accuse the same healthy workload; the honest agents see nothing wrong. *Middle (liveness):* k agents go silent (equivalent to crashed or refusing to vote) while a real attack happens. *Right (deception):* k liars accuse a healthy workload that really does show a weak blip to exactly one honest agent, the case where an honest agent can be tipped into voting. k = 3 is shown on the left only, beyond the theory's limit, to confirm the quorum is the only barrier. **What the data shows.**

- **Safety, accusation only (left):** 0, 1 and 2 liars never isolated the healthy workload (0/20 each; the honest agents do not vote without seeing the anomaly themselves); with **3** liars it happened in 20/20 (they alone make a valid certificate). So pure accusation is safe up to 2 compromised agents, one better than the theory's minimum.
- **Liveness (middle):** with 0 or 1 silent agents the attack was contained 20/20; with **2** silent agents **0/20**: only two votes were ever collected (the table's "most votes seen" is 2). This is the BFT bound showing exactly where theory predicts, for a crash-like fault.
- **Deception (right):** with 0 or 1 liars nothing was isolated (0/20), but with **2 liars plus one honest agent that really sees a weak blip** the healthy workload was isolated in **20/20**: the two liars' signatures plus the one deceived honest signature make three. **So 2 compromised agents out of 4 do break safety**, as theory says, though it needs a real anomaly for the honest agent to be deceived by.

Honest framing: the platform's guarantee is "1 compromised agent is tolerated"; at 2 it fails in two of the three ways tested. The tests are designed to find the breaking point, so they should be read as a boundary, not typical behaviour.

4.6 Agent crash SIMULATOR

Agent crash: 4, 3 and 2 of 4 agents alive



Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

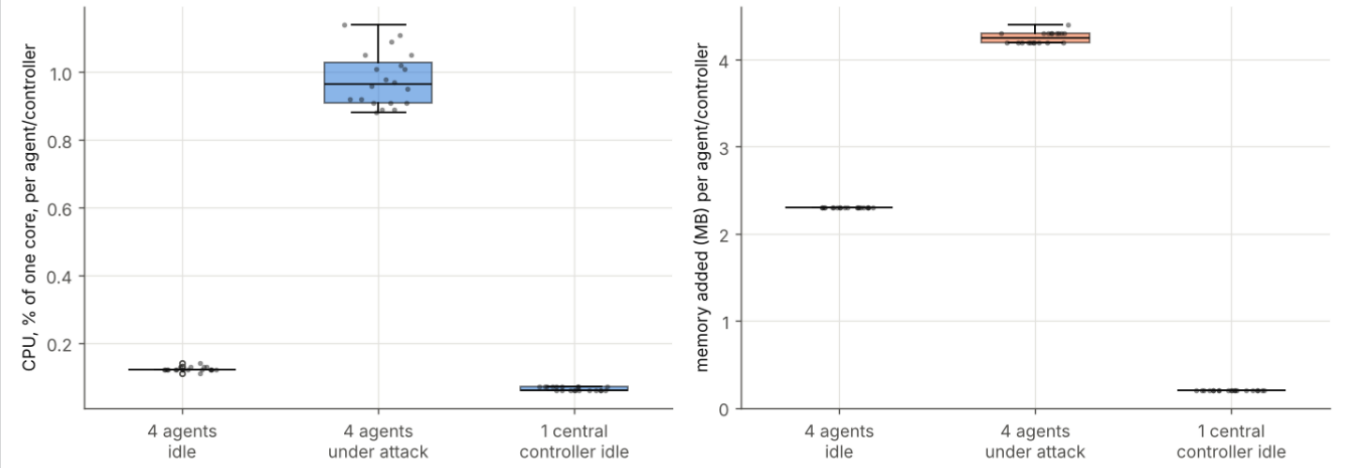
Agents alive	Trials	Attack contained (30 s)	Full cycle completed	Time to isolate (s)	Time to full access (s)	Most CONTAIN votes seen	Other workload wrongly isolated
4	20	20/20 = 100% (95% CI 84-100%)	20/20 = 100% (95% CI 84-100%)	median 1.1 (IQR 1.1-1.2, min 1.1, max 1.6, n=20)	median 18.5 (IQR 18.5-18.5, min 18.5, max 19.0, n=20)	0	0/20
3	20	20/20 = 100% (95% CI 84-100%)	20/20 = 100% (95% CI 84-100%)	median 1.5 (IQR 1.0-1.8, min 1.0, max 3.0, n=20)	median 19.0 (IQR 19.0-20.0, min 18.5, max 22.0, n=20)	0	0/20
2	20	0/20 = 0% (95% CI 0-16%)	0/20 = 0% (95% CI 0-16%)	n/a (no trial reached it)	n/a (no trial reached it)	2	0/20

How to read it. A real attack with 4, 3 and 2 agents alive. With 3 alive the three survivors can still form the 3-signature quorum and finish the whole pipeline. With 2 alive there are not enough signatures, so nothing is isolated; the table shows how many votes the survivors collected. **What the data shows.** With 4 or 3 agents alive the attack was contained and the whole cycle completed in 20/20 trials (isolation median 1.1 s with 4, 1.5 s with 3; full access 18.5 s and 19.0 s). With **2** alive nothing was isolated (0/20): the survivors collected 2 votes and never the 3 needed. Importantly it **fails safe**: no healthy workload was isolated either (0/20). The cost is that during that time a genuine attack is not contained automatically; the dashboard shows the proposal waiting for votes, and a human has to act.

4.7 Agent overhead SIMULATOR

Per-agent CPU and memory in the simulator (indicative only)

Dashboard Part 4: Evaluation tab, full page (before Phase C/D changes) (part 5/6)

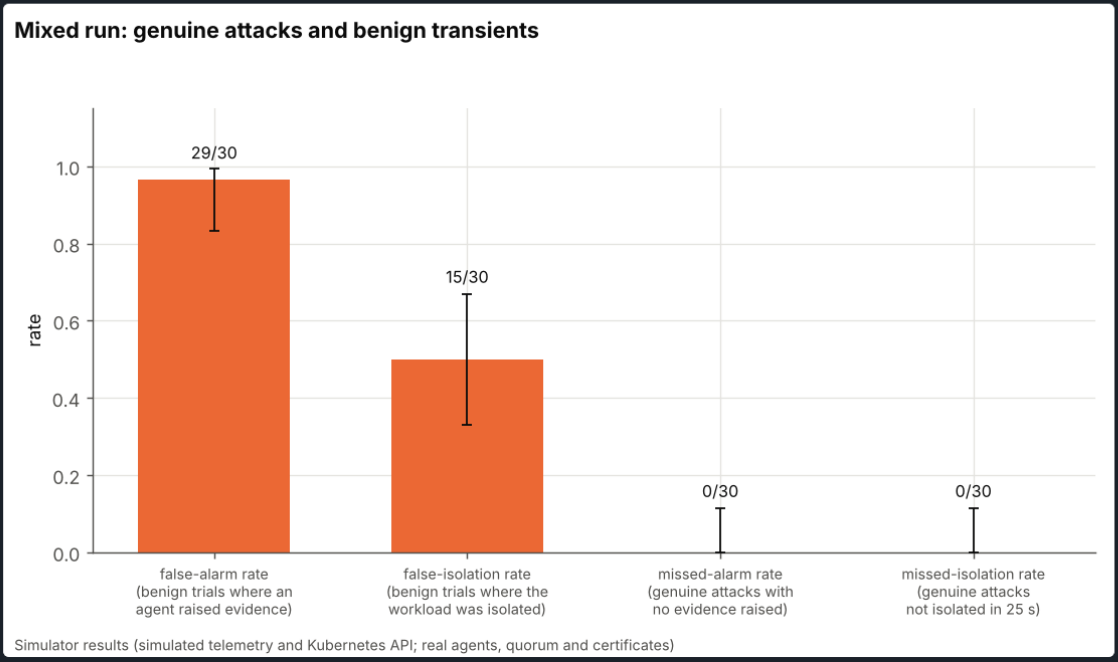


Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Configuration	Trials	CPU, % of one core, per unit	Memory added per unit (MB)	Whole process RSS (MB)
4 agents, idle	20	median 0.1 (IQR 0.1-0.1, min 0.1, max 0.1, n=20)	median 2.3 (IQR 2.3-2.3, min 2.3, max 2.3, n=20)	median 58.7 (IQR 58.7-58.8, min 58.6, max 58.8, n=20)
4 agents, during an attack + recovery	20	median 1.0 (IQR 0.9-1.0, min 0.9, max 1.1, n=20)	median 4.2 (IQR 4.2-4.3, min 4.2, max 4.4, n=20)	median 66.7 (IQR 66.5-66.8, min 66.3, max 67.1, n=20)
1 central controller, idle	20	median 0.1 (IQR 0.1-0.1, min 0.1, max 0.1, n=20)	median 0.2 (IQR 0.2-0.2, min 0.2, max 0.2, n=20)	median 49.8 (IQR 49.7-49.8, min 49.7, max 49.9, n=20)

How to read it, and why to be careful. In the simulator all four agents (plus their gRPC servers and the simulated telemetry) share **one** Python process, so these are the process's CPU and memory **divided by four**. Real agents run in their own pods, poll real HTTP endpoints every second and talk to a real API server, so real numbers will differ (and each pod has the interpreter's baseline memory, which this division does not capture). Treat the figure as "the agent logic is cheap", not as a sizing guide. For real numbers use `kubectl top pods -n resilience` on your cluster.
What the data shows. Per agent, in the simulator: about 0.1% of one core when idle and about 1.0% during an attack and recovery; about 2.3 MB of memory added when idle and 4.2 MB during an attack. The single central controller is lighter (0.1% CPU, 0.2 MB) since it has no gRPC servers, signatures or votes. These are tiny because the simulated telemetry is free; the real agent's HTTP polling of four workloads every second will cost more. This supports "the decision logic is cheap" and nothing more.

4.8 False positives and false negatives in a mixed run SIMULATOR



Simulator results (simulated telemetry and Kubernetes API; real agents, quorum and certificates)

Level	Quantity	Result
Alarm (evidence signed above 0.3 confidence)	false-alarm rate = FP / benign trials	29/30 = 97% (95% CI 83-99%)
Alarm	missed-alarm rate (false negative) = FN / genuine trials	0/30 = 0% (95% CI 0-11%)
Isolation (quorum committed)	false-isolation rate = FP / benign trials	15/30 = 50% (95% CI 33-67%)
Isolation	missed-isolation rate (false negative) = FN / genuine trials	0/30 = 0% (95% CI 0-11%)
Isolation	precision = TP / (TP + FP)	67%
Isolation	recall = TP / genuine trials	30/30 = 100% (95% CI 89-100%)

Benign transient strength	Seen by	Trials	False alarm	False isolation
at or below the alert threshold (<= 8 connections)	1-2 agents	1	0/1	0/1
moderate (9-16 connections, confidence 0.53-0.75)	1-2 agents	3	3/3	0/3
moderate (9-16 connections, confidence 0.53-0.75)	3-4 agents	14	14/14	11/14
strong (> 16 connections, confidence > 0.75)	1-2 agents	8	8/8	0/8
strong (> 16 connections, confidence > 0.75)	3-4 agents	4	4/4	4/4

How to read it. 30 genuine attacks (random mixes of the three attack signals, random target) and 30 benign transients (a short spike in outbound

Dashboard Part 4: Evaluation tab, full page (before Phase C/D changes) (part 6/6)

connections of random strength, seen by 1-4 agents, or a long weak burst seen by all) are run through the full platform. The four bars are rates with 95% confidence intervals. **These rates are only as meaningful as the benign mix**: the benign transients here were *chosen* to straddle the detection thresholds (the connection rule rates nothing up to 8 connections, then 0.5 at 9, rising to 1.0 at 24; the normal level is about 2), so some are harmless and some look exactly like an attack. A production environment with calmer or noisier benign traffic would give different rates. The table that splits benign trials by strength shows where the line is. **What the data shows.** All 30 genuine attacks raised an alarm and were isolated (0/30 missed, 95% CI 0-11%). Of the 30 benign transients 29 raised an alarm (a false-alarm rate of 97%, which is expected: almost all were above the 8-connection threshold by construction) and **15 (50%, CI 33-67%) led to an isolation**. Precision at isolation level was 67%. The stratified table shows what decides it: **no benign transient was ever isolated when only 1-2 agents saw it (0 of 12 trials), and 15 of 18 were isolated when 3-4 agents saw it** (all four of the strong ones, 11 of 14 of the moderate ones; the 3 that were not were all at the weak end, 9.8-11.9 connections). In other words the platform's protection against false isolation is that **a benign disturbance has to be visible to three agents at once**. A transient that all agents see above the threshold is, to a threshold detector, indistinguishable from an attack. In the simulator partial visibility is a modelling choice (the `observers` set); in a real cluster all four agents poll the same endpoints, so most real spikes would be visible to all of them, and the 50% should be read as a stress-test bound for a deliberately hostile benign mix, not a prediction. This is the platform's main accuracy limit (Phase 2's investigation helps weak or disputed evidence, not unanimous ones).

Honest limitations of this evaluation

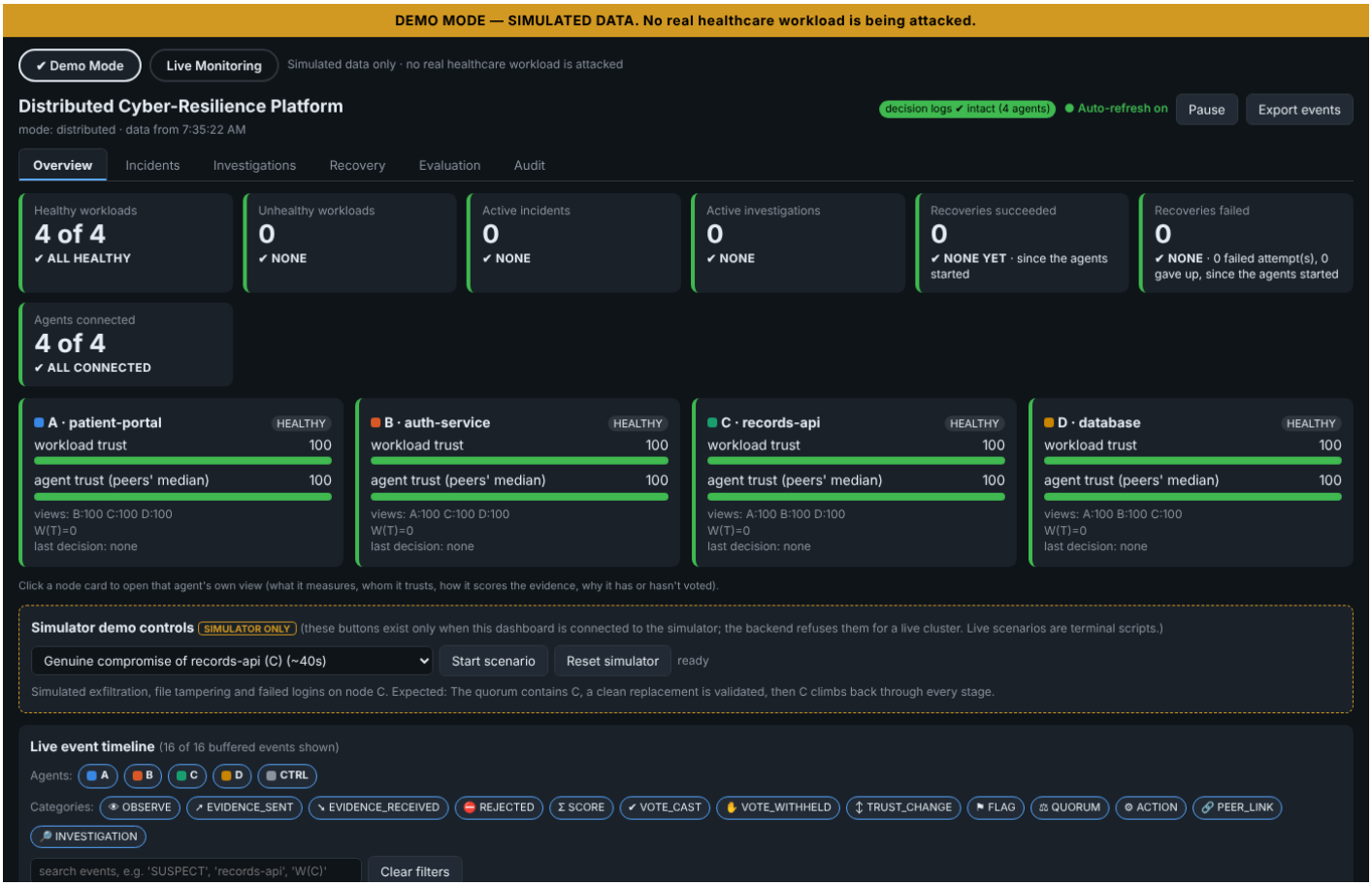
- **Simulated telemetry.** A "genuine attack" is a simulated signal pattern (outbound connections, a modified file hash, a suspicious process, failed logins). Nothing real is attacked or made to misbehave. The detector is a threshold detector, so any benign input beyond its thresholds is by construction indistinguishable from an attack at the single-agent level; what the platform adds is the quorum and the investigation.
- **Benign and attack mixes are my choice**, so the false-positive/negative rates are illustrations of behaviour under a stated mix, not estimates of real-world rates.
- **Seeds fix inputs, not timings** (section 1). With 20 trials per variant, small differences between medians are noise; differences are only worth quoting where the confidence intervals or interquartile ranges do not overlap.
- **No real Kubernetes:** no real pod start times, no Calico enforcement, no API latency; those are what `scripts/verify-isolation.sh`, `verify-rbac.sh` and `verify-webhook.sh` are for.
- **Silent agents are crashed agents** (the simulator has no "stay alive but refuse to vote" mode); for the liveness question they are equivalent.
- Parallel trials share one machine; the CPU/memory experiment is serial for that reason, but it is still a single-process approximation.
- The 2-of-4 safety break shown in the "deceived" test is a property of any 3-of-4 quorum, not a bug: see `docs/SECURITY.md` section 1.

▸ **What was run, and how to trust it**

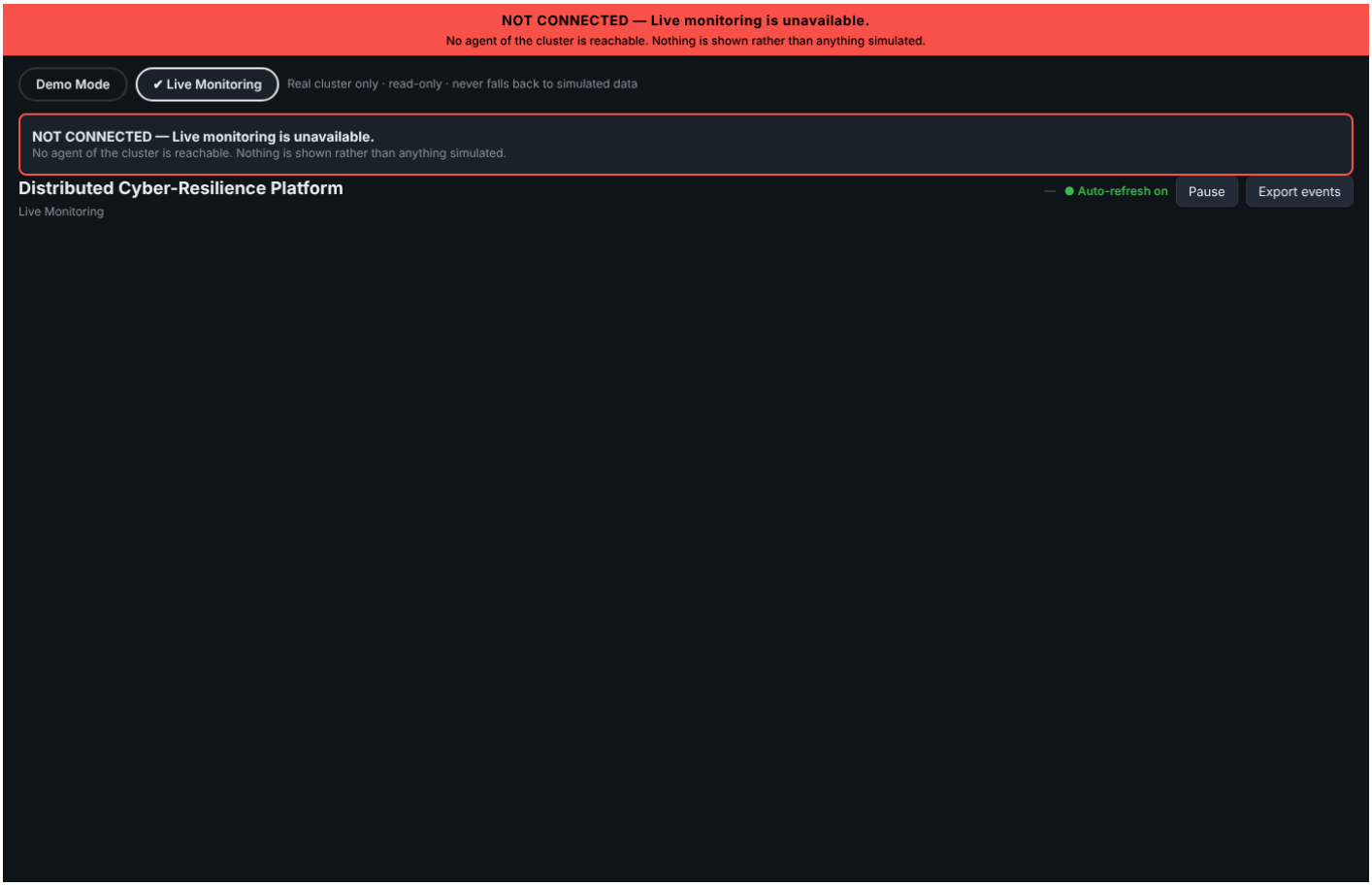
▸ **Definitions (start event, end event, denominator)**

▸ **The experiments in one paragraph each**

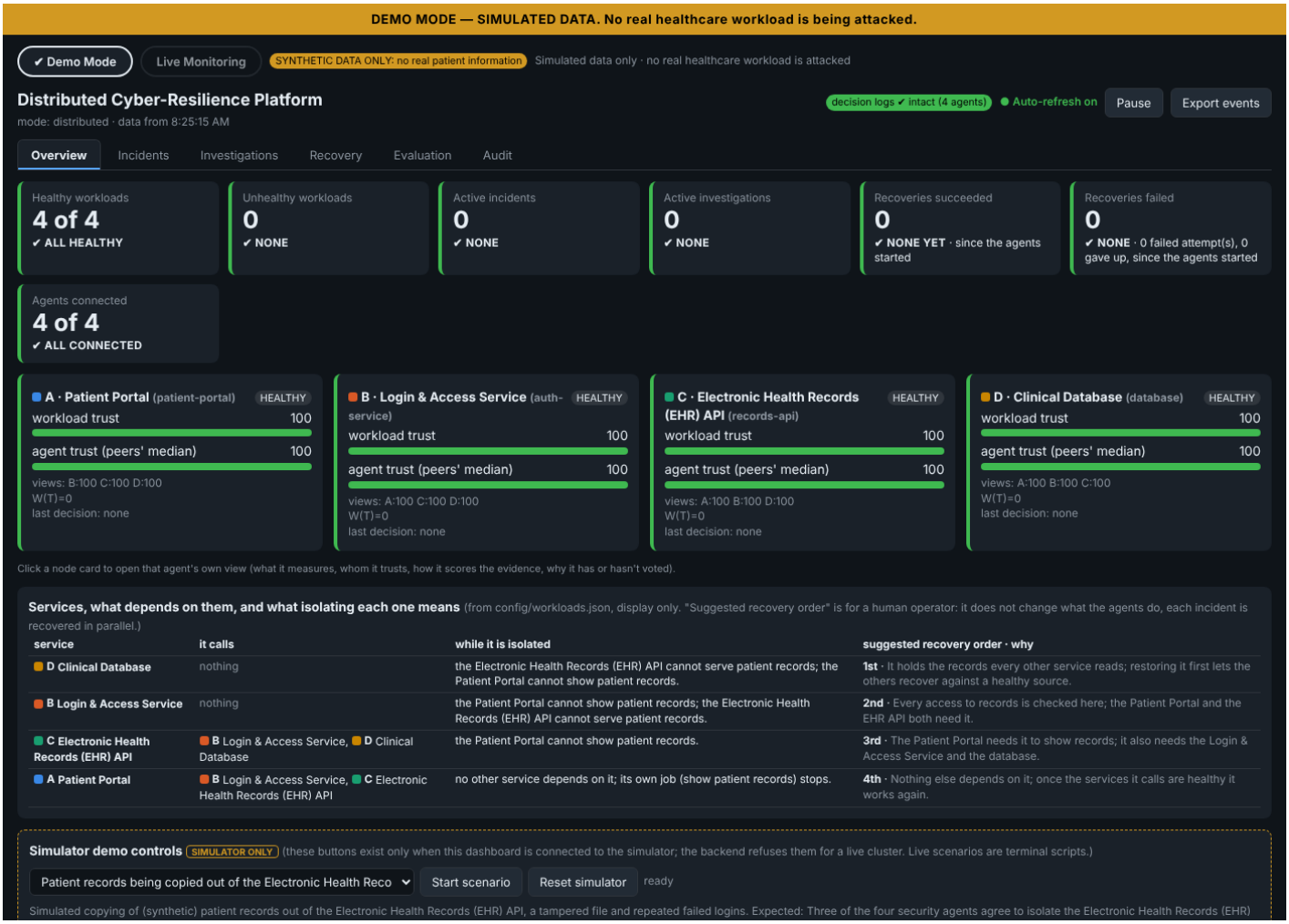
Healthcare Phase B: Demo Mode



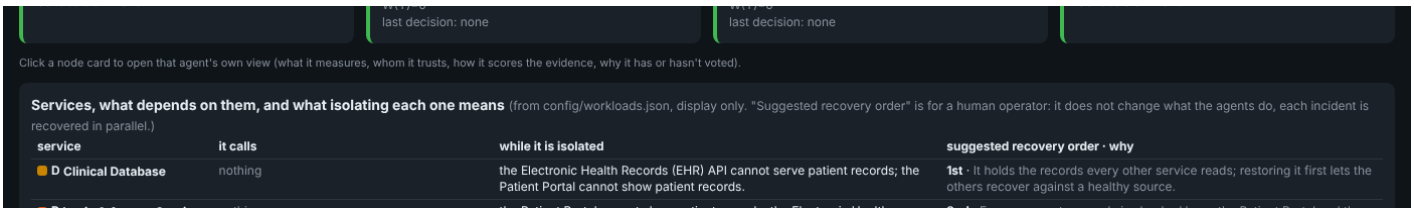
Healthcare Phase B: Live Monitoring - NOT CONNECTED (real behaviour, no cluster)



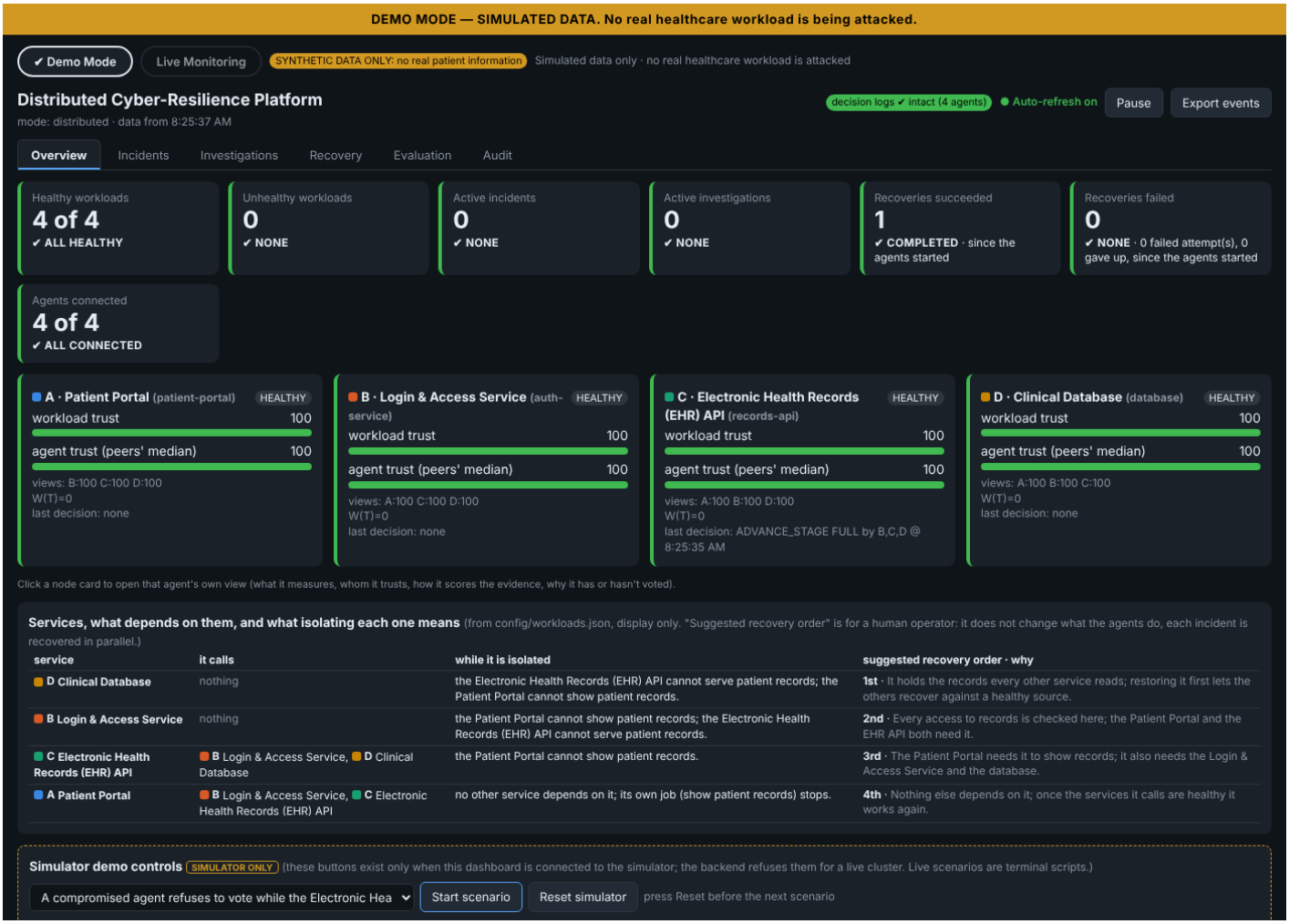
Healthcare Phase C: healthcare names and Services panel



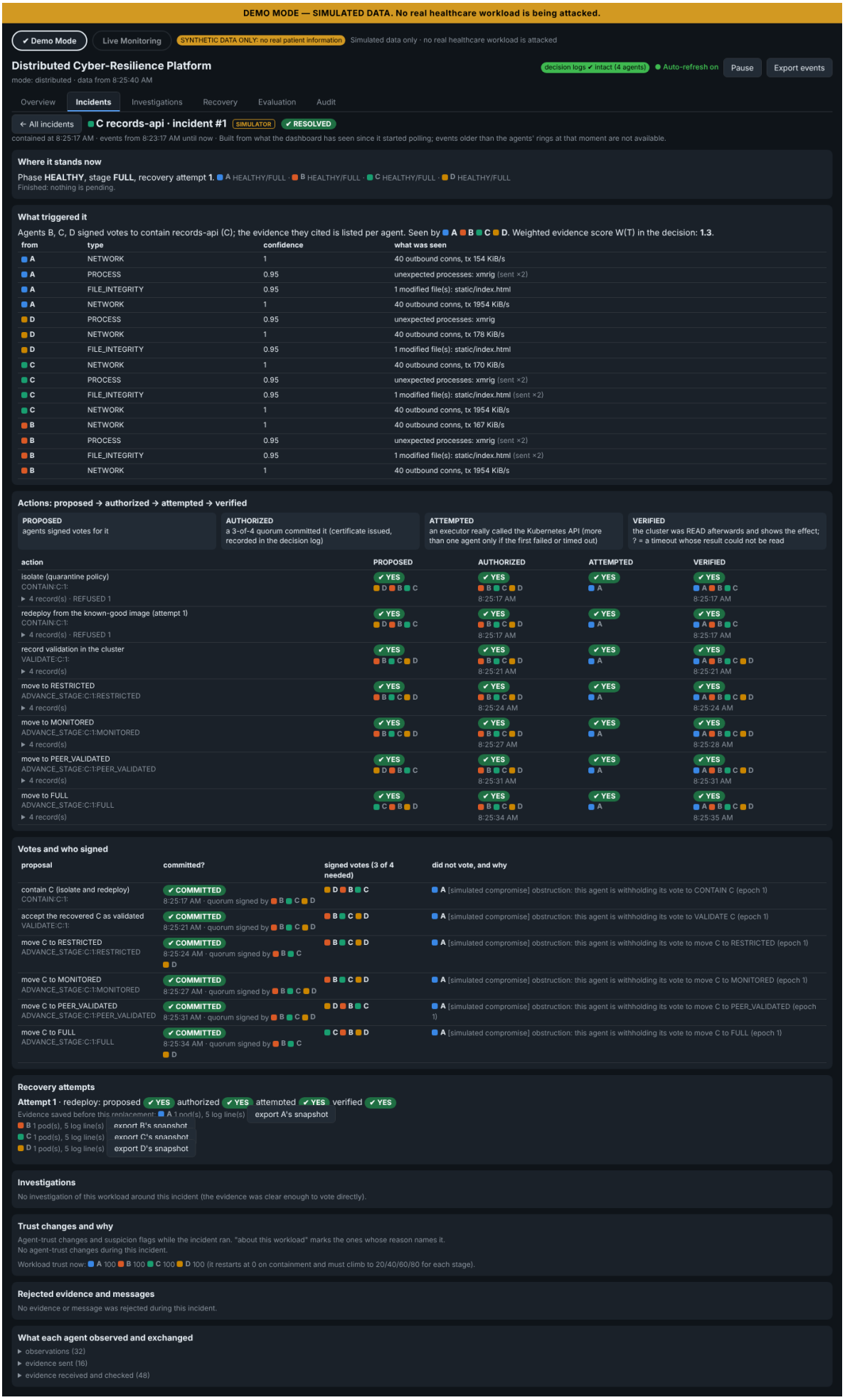
Healthcare Phase C: scenario picker with the 'obstruct' scenario



Healthcare Phase C: obstruct scenario finished (PASS)



Healthcare Phase C: incident view of the obstruct scenario



DEMO MODE — SIMULATED DATA. No real healthcare workload is being attacked.

✔ Demo Mode | Live Monitoring | SYNTHETIC DATA ONLY: no real patient information | Simulated data only · no real healthcare workload is attacked

Distributed Cyber-Resilience Platform

mode: distributed · data from 8:25:40 AM

decision logs ✔ intact (4 agents) | Auto-refresh on | Pause | Export events

Overview | Incidents | Investigations | Recovery | Evaluation | Audit

← All incidents

C records-api · incident #1 SIMULATOR ✔ RESOLVED

contained at 8:25:17 AM · events from 8:23:17 AM until now · Built from what the dashboard has seen since it started polling; events older than the agents' rings at that moment are not available.

Where it stands now

Phase **HEALTHY**, stage **FULL**, recovery attempt **1**. A HEALTHY/FULL · B HEALTHY/FULL · C HEALTHY/FULL · D HEALTHY/FULL

Finished: nothing is pending.

What triggered it

Agents B, C, D signed votes to contain records-api (C); the evidence they cited is listed per agent. Seen by **A B C D**. Weighted evidence score W(T) in the decision: **1.3**.

from	type	confidence	what was seen
A	NETWORK	1	40 outbound conns, tx 154 KiB/s
A	PROCESS	0.95	unexpected processes: xmrig (sent ×2)
A	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
A	NETWORK	1	40 outbound conns, tx 1954 KiB/s
D	PROCESS	0.95	unexpected processes: xmrig
D	NETWORK	1	40 outbound conns, tx 178 KiB/s
D	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html
C	NETWORK	1	40 outbound conns, tx 170 KiB/s
C	PROCESS	0.95	unexpected processes: xmrig (sent ×2)
C	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html (sent ×2)
C	NETWORK	1	40 outbound conns, tx 1954 KiB/s
B	NETWORK	1	40 outbound conns, tx 167 KiB/s
B	PROCESS	0.95	unexpected processes: xmrig (sent ×2)
B	FILE_INTEGRITY	0.95	1 modified file(s): static/index.html (sent ×2)
B	NETWORK	1	40 outbound conns, tx 1954 KiB/s

Actions: proposed → authorized → attempted → verified

PROPOSED agents signed votes for it

AUTHORIZED a 3-of-4 quorum committed it (certificate issued, recorded in the decision log)

ATTEMPTED an executor really called the Kubernetes API (more than one agent only if the first failed or timed out)

VERIFIED the cluster was READ afterwards and shows the effect; ? = a timeout whose result could not be read

action	PROPOSED	AUTHORIZED	ATTEMPTED	VERIFIED
isolate (quarantine policy) CONTAIN:C:1: ▸ 4 record(s) · REFUSED 1	✔ YES D B C	✔ YES B C D 8:25:17 AM	✔ YES A	✔ YES A B C 8:25:17 AM
redeploy from the known-good image (attempt 1) CONTAIN:C:1: ▸ 4 record(s) · REFUSED 1	✔ YES D B C	✔ YES B C D 8:25:17 AM	✔ YES A	✔ YES A B C 8:25:17 AM
record validation in the cluster VALIDATE:C:1: ▸ 4 record(s)	✔ YES B C D	✔ YES B C D 8:25:21 AM	✔ YES A	✔ YES A B C D 8:25:21 AM
move to RESTRICTED ADVANCE_STAGE:C:1:RESTRICTED ▸ 4 record(s)	✔ YES B C D	✔ YES B C D 8:25:24 AM	✔ YES A	✔ YES A B C D 8:25:24 AM
move to MONITORED ADVANCE_STAGE:C:1:MONITORED ▸ 4 record(s)	✔ YES B C D	✔ YES B C D 8:25:27 AM	✔ YES A	✔ YES A B C D 8:25:28 AM
move to PEER_VALIDATED ADVANCE_STAGE:C:1:PEER_VALIDATED ▸ 4 record(s)	✔ YES D B C	✔ YES B C D 8:25:31 AM	✔ YES A	✔ YES A B C D 8:25:31 AM
move to FULL ADVANCE_STAGE:C:1:FULL ▸ 4 record(s)	✔ YES C B D	✔ YES B C D 8:25:34 AM	✔ YES A	✔ YES A B C D 8:25:35 AM

Votes and who signed

proposal	committed?	signed votes (3 of 4 needed)	did not vote, and why
contain C (isolate and redeploy) CONTAIN:C:1:	✔ COMMITTED 8:25:17 AM · quorum signed by B C D	D B C	A [simulated compromise] obstruction: this agent is withholding its vote to CONTAIN C (epoch 1)
accept the recovered C as validated VALIDATE:C:1:	✔ COMMITTED 8:25:21 AM · quorum signed by B C D	B C D	A [simulated compromise] obstruction: this agent is withholding its vote to VALIDATE C (epoch 1)
move C to RESTRICTED ADVANCE_STAGE:C:1:RESTRICTED	✔ COMMITTED 8:25:24 AM · quorum signed by B C D	B C D	A [simulated compromise] obstruction: this agent is withholding its vote to move C to RESTRICTED (epoch 1)
move C to MONITORED ADVANCE_STAGE:C:1:MONITORED	✔ COMMITTED 8:25:27 AM · quorum signed by B C D	B C D	A [simulated compromise] obstruction: this agent is withholding its vote to move C to MONITORED (epoch 1)
move C to PEER_VALIDATED ADVANCE_STAGE:C:1:PEER_VALIDATED	✔ COMMITTED 8:25:31 AM · quorum signed by B C D	D B C	A [simulated compromise] obstruction: this agent is withholding its vote to move C to PEER_VALIDATED (epoch 1)
move C to FULL ADVANCE_STAGE:C:1:FULL	✔ COMMITTED 8:25:34 AM · quorum signed by B C D	C B D	A [simulated compromise] obstruction: this agent is withholding its vote to move C to FULL (epoch 1)

Recovery attempts

Attempt 1 · redeploy: proposed ✔ YES authorized ✔ YES attempted ✔ YES verified ✔ YES

Evidence saved before this replacement: A 1 pod(s), 5 log line(s) export A's snapshot

B 1 pod(s), 5 log line(s) export B's snapshot

C 1 pod(s), 5 log line(s) export C's snapshot

D 1 pod(s), 5 log line(s) export D's snapshot

Investigations

No investigation of this workload around this incident (the evidence was clear enough to vote directly).

Trust changes and why

Agent-trust changes and suspicion flags while the incident ran. "about this workload" marks the ones whose reason names it.

No agent-trust changes during this incident.

Workload trust now: A 100 B 100 C 100 D 100 (it restarts at 0 on containment and must climb to 20/40/60/80 for each stage).

Rejected evidence and messages

No evidence or message was rejected during this incident.

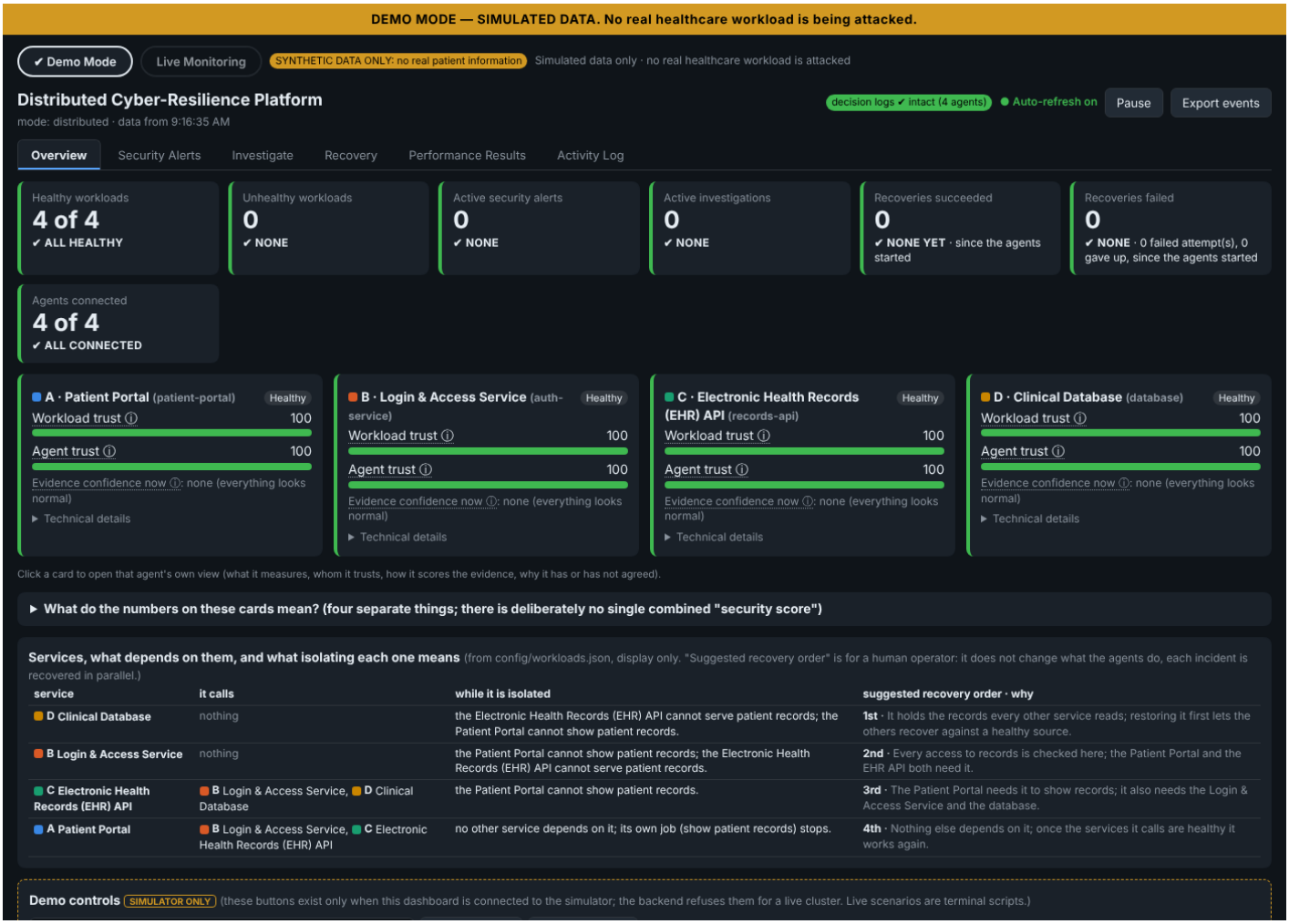
What each agent observed and exchanged

▸ observations (32)

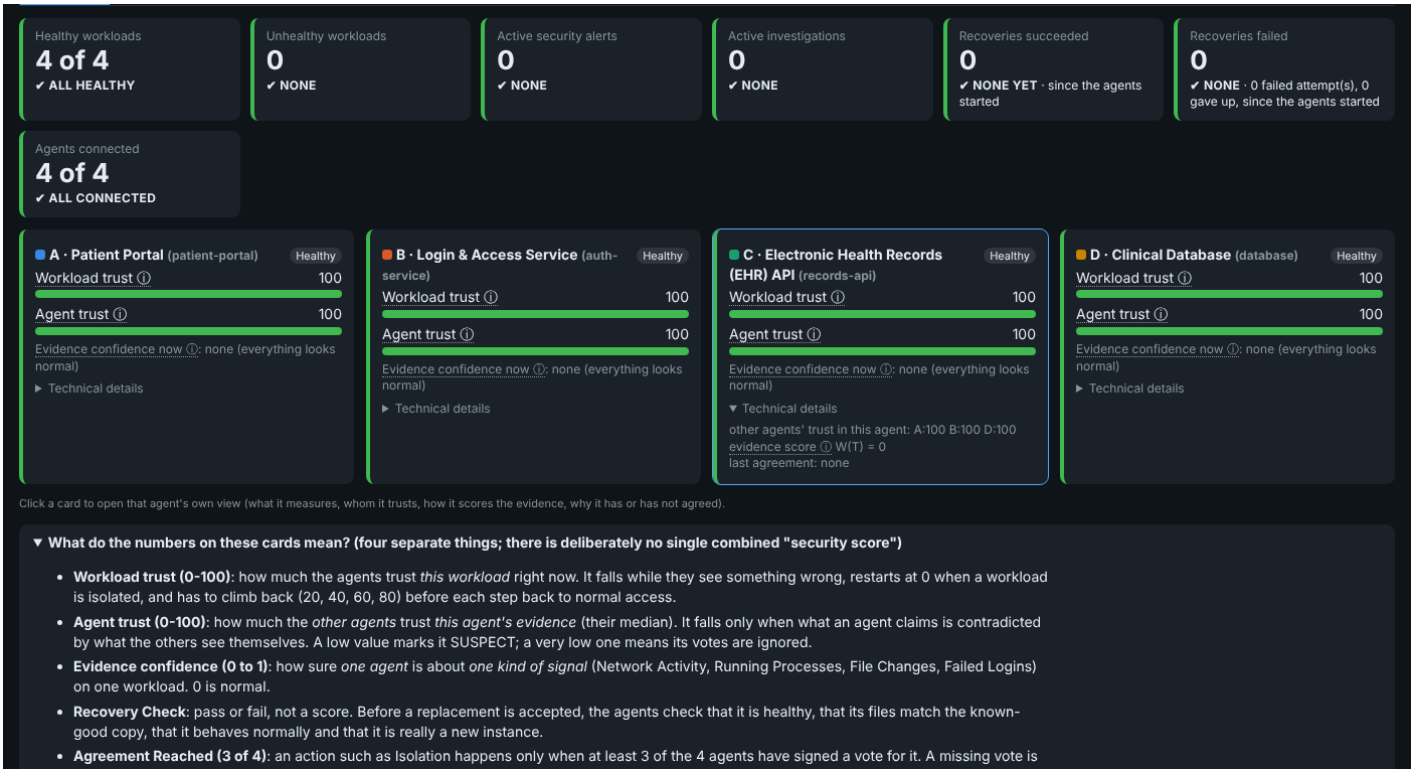
▸ evidence sent (16)

▸ evidence received and checked (48)

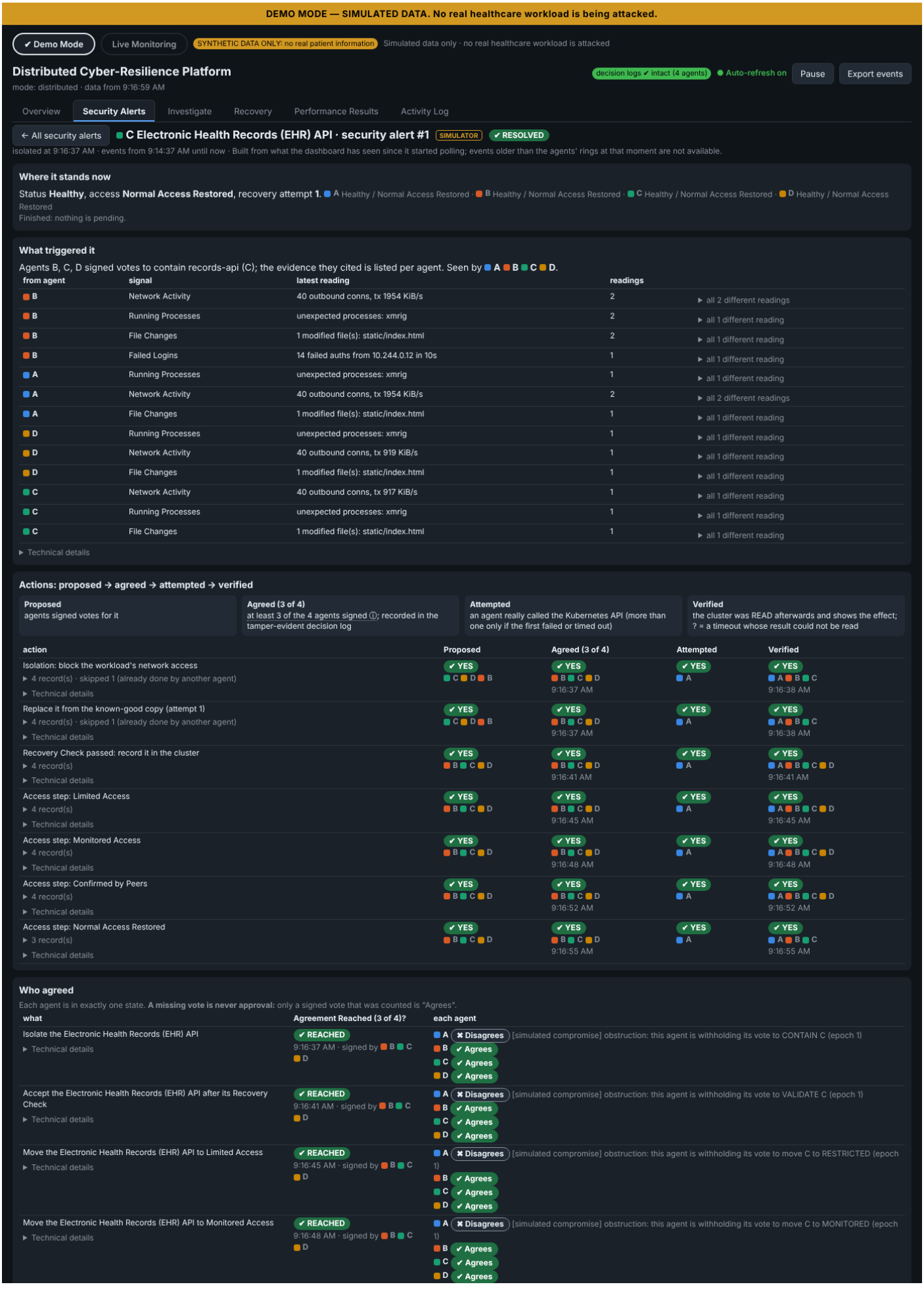
Healthcare Phase D: plain-English Overview



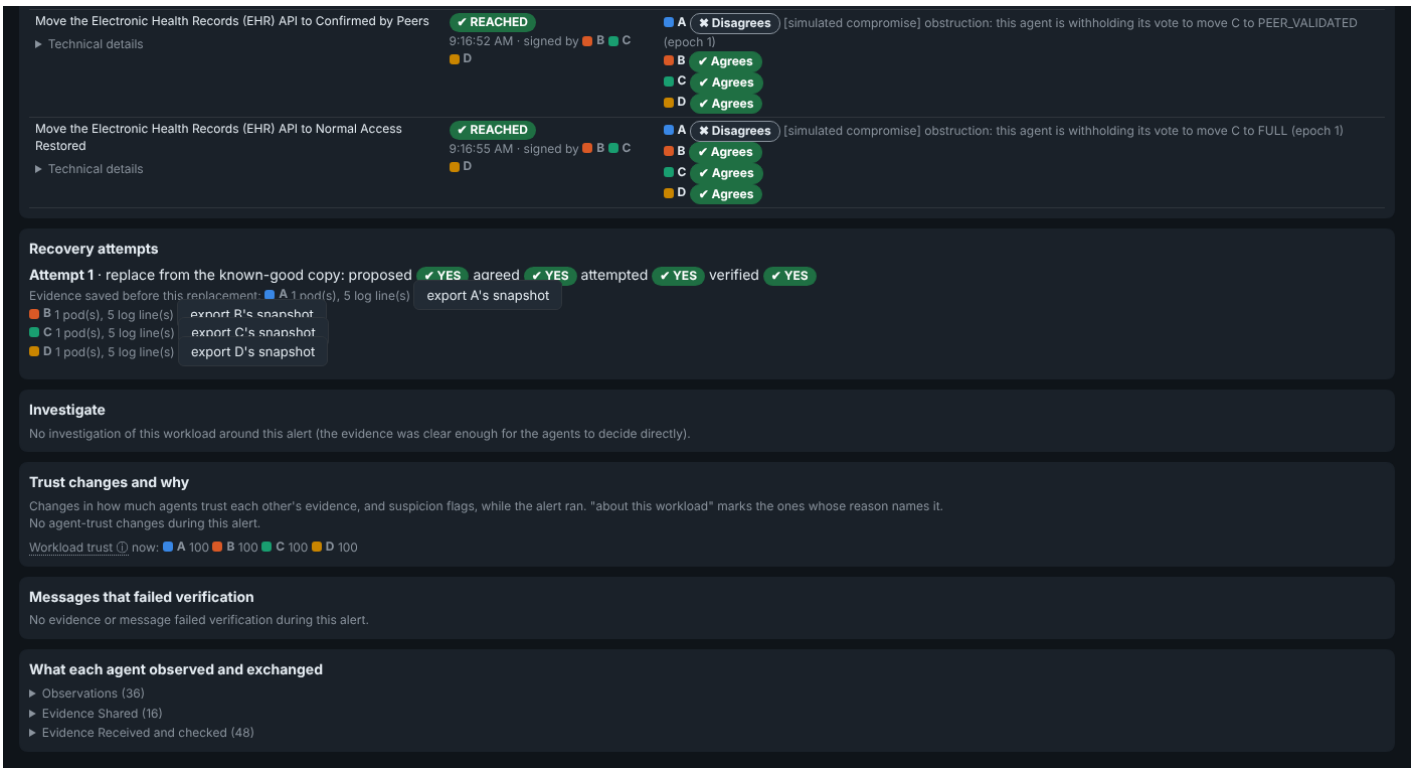
Healthcare Phase D: explanations and technical details



Healthcare Phase D: security alert, grouped evidence, five agent states (part 1/2)



Healthcare Phase D: security alert, grouped evidence, five agent states (part 2/2)



Healthcare Phase D: Performance Results with collapsed explanations

